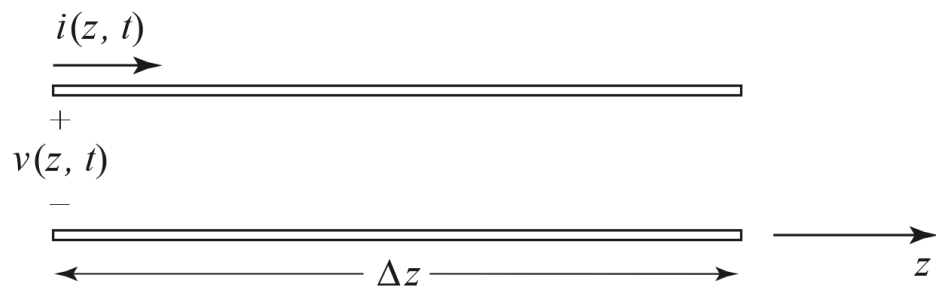


微波电路原理与设计

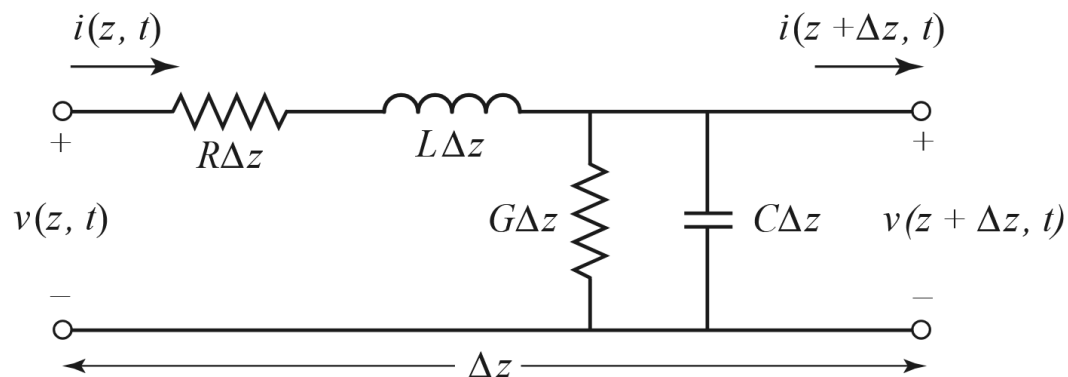
第一章传输线理论

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2026-03

传输线的物理模型



(a)



(b)

KVL

$$v(z, t) - R\Delta z i(z, t) - L\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0,$$

KCL

$$i(z, t) - G\Delta z v(z + \Delta z, t) - C\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0.$$

$\Delta z \rightarrow 0$ ↓

$$\frac{\partial v(z, t)}{\partial z} = -Ri(z, t) - L \frac{\partial i(z, t)}{\partial t},$$

$$\frac{\partial i(z, t)}{\partial z} = -Gv(z, t) - C \frac{\partial v(z, t)}{\partial t}.$$

↓

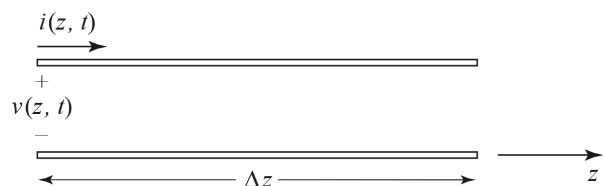
$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\}$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\}$$

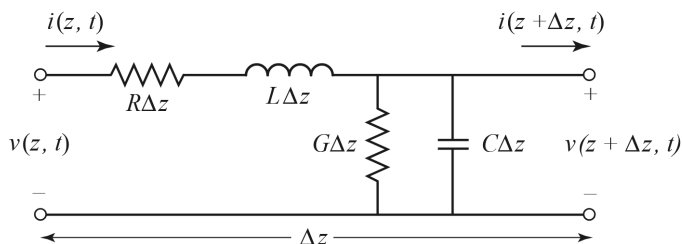
$$\frac{dV(z)}{dz} = -(R + j\omega L)I(z),$$

$$\frac{dI(z)}{dz} = -(G + j\omega C)V(z).$$

传输线模型



(a)



(b)

$$\frac{d^2 V(z)}{dz^2} - \gamma^2 V(z) = 0,$$

$$\frac{d^2 I(z)}{dz^2} - \gamma^2 I(z) = 0,$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

正向传输 反向传输

电压波 $V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z},$

电流波 $I(z) = I_o^+ e^{-\gamma z} + I_o^- e^{\gamma z},$

$$I(z) = \frac{\gamma}{R + j\omega L} (V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}). \quad Z_0 = \frac{R + j\omega L}{\gamma} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \quad \frac{V_o^+}{I_o^+} = Z_0 = \frac{-V_o^-}{I_o^-}. \quad I(z) = \frac{V_o^+}{Z_0} e^{-\gamma z} - \frac{V_o^-}{Z_0} e^{\gamma z}.$$

无耗传输线方程:

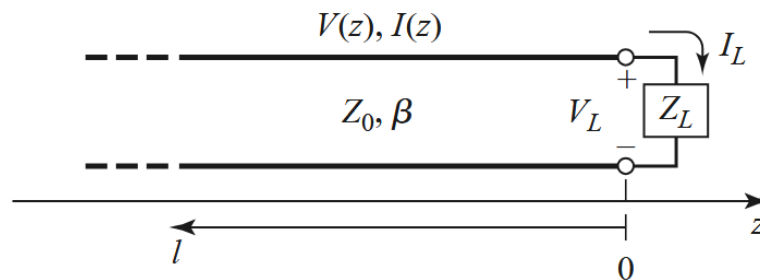
$$V(z) = V_o^+ e^{-j\beta z} + V_o^- e^{j\beta z},$$

$$I(z) = \frac{V_o^+}{Z_0} e^{-j\beta z} - \frac{V_o^-}{Z_0} e^{j\beta z}.$$

$$\beta = \omega \sqrt{LC} = \frac{2\pi}{\lambda}$$

自由空间中的波长

终端加载的无耗传输线



$$V(z) = V_o^+ e^{-j\beta z} + V_o^- e^{j\beta z}.$$

$$I(z) = \frac{V_o^+}{Z_0} e^{-j\beta z} - \frac{V_o^-}{Z_0} e^{j\beta z}.$$

$$Z_L = \frac{V(0)}{I(0)} = \frac{V_o^+ + V_o^-}{V_o^+ - V_o^-} Z_0.$$

反射系数:

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

考虑反射系数后的表达式:

$$V(z) = V_o^+ (e^{-j\beta z} + \Gamma e^{j\beta z}),$$

$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}).$$

$$\begin{aligned} \Gamma(-z) &= \Gamma(l) \\ &= \Gamma e^{2j\beta z} \\ &= \Gamma e^{-2j\beta l} \end{aligned}$$

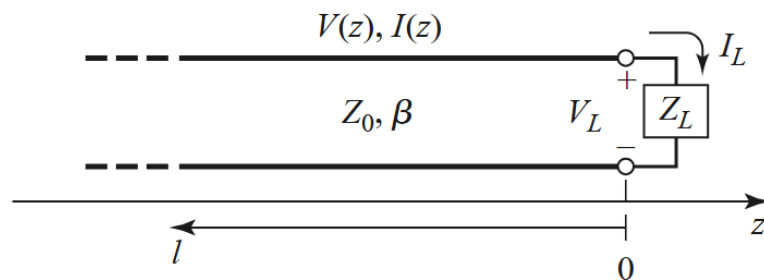
功率关系:

$$P_{\text{avg}} = \frac{1}{2} \text{Re}\{V(z)I(z)^*\} = \frac{1}{2} \frac{|V_o^+|^2}{Z_0} \text{Re}\{1 - \Gamma^* e^{-2j\beta z} + \Gamma e^{2j\beta z} - |\Gamma|^2\}, = \frac{1}{2} \frac{|V_o^+|^2}{Z_0} (1 - |\Gamma|^2)$$

这表明在传输线上的任意位置, **平均功率流都是恒定的**。因此, 传递到负载的总平均功率 P_{avg} 等于**入射功率** ($|V_o^+|^2/2Z_0$) **减去反射功率** ($|V_o^+|^2|\Gamma|^2/2Z_0$)。

当 $\Gamma = 0$ 时, **最大功率被传输到负载**; 而当 $|\Gamma| = 1$ 时, **没有功率传输到负载**。

终端加载的无耗传输线



考虑反射系数后的表达式:

$$V(z) = V_o^+ (e^{-j\beta z} + \Gamma e^{j\beta z}),$$

$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}).$$

$$\begin{aligned}
 |V(z)| &= |V_o^+| |1 + \Gamma e^{2j\beta z}| = |V_o^+| |1 + \Gamma e^{-2j\beta \ell}| \\
 &= |V_o^+| |1 + |\Gamma| e^{j(\theta - 2\beta \ell)}|,
 \end{aligned}$$

$$\Gamma = |\Gamma| e^{j\theta}$$

$$V_{\max} = |V_o^+| (1 + |\Gamma|) \quad V_{\min} = |V_o^+| (1 - |\Gamma|)$$

驻波比 (Standing wave ratio, SWR) 或电压驻波比 (Voltage SWR, VSWR)

$$\rho = \text{SWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

反射系数及dB

dB表示为分贝（相对值），通常表示信号强度

$$dB = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{|V_2|^2}{|V_1|^2} = 20 \log_{10} \frac{|V_2|}{|V_1|}$$

反射系数以dB表示 $\Gamma(dB) = 10 \log_{10} |\Gamma|^2$

回波损耗 $RL(dB) = -10 \log_{10} |\Gamma|^2$

- dBm、dBw表示为分贝毫瓦、瓦（相对1mW、1W而言）（功率单位）

$$P_T(dBm) = 10 \log_{10} \frac{P_T}{1mW} \quad P_T(dBW) = 10 \log_{10} \frac{P_T}{1W}$$

反射系数 Γ	回波损耗RL	驻波比VSWR
1.00	0.00	∞
0.90	0.92	19.00
0.80	0.94	9.00
0.70	3.10	5.67
0.60	4.44	4.00
0.50	6.02	3.00
0.40	7.96	2.33
0.30	10.46	1.86
0.20	13.98	1.50
0.10	20.00	1.22
0.09	20.92	1.20
0.08	21.94	1.17
0.07	23.10	1.15
0.06	24.44	1.13
0.05	26.02	1.11
0.04	27.96	1.08
0.03	30.46	1.06
0.02	33.98	1.04
0.01	40.00	1.02
0.00	∞	1.00

知识点

$$a(dB) = 10 \log_{10} a$$

$$2a(dB) = 10 \log_{10} (2a) = 10 \log_{10} 2 + 10 \log_{10} a = \boxed{3(dB)} + a(dB)$$

$$X(dB) + 3dB \rightarrow X * 2$$

$$\text{同理 } X(dB) - 3dB \rightarrow X * 1/2$$

$$a(dB) = 10 \log_{10} a$$

$$10a(dB) = 10 \log_{10} (10a) = 10 \log_{10} 10 + 10 \log_{10} a = \boxed{10(dB)} + a(dB)$$

$$X(dB) + 10dB \rightarrow X * 10$$

$$\text{同理 } X(dB) - 10dB \rightarrow X * 1/10$$

知识点

$$P_T|_{\text{dBm}} = 10 \cdot \log_{10} \left(\frac{P_T}{1 \text{ mW}} \right)$$

$$P_T|_{\text{dBW}} = 10 \cdot \log_{10} \left(\frac{P_T}{1 \text{ W}} \right) = P_T|_{\text{dBm}} - 30 \text{ dB}$$

$$\begin{aligned} \frac{P_1}{P_2} &\triangleq 10 \cdot \log_{10} \left(\frac{P_1}{P_2} \right) \text{ dB} = 10 \cdot \log_{10} \left(\frac{P_1}{1 \text{ W}} \right) - 10 \cdot \log_{10} \left(\frac{P_2}{1 \text{ W}} \right) = P_1|_{\text{dBW}} - P_2|_{\text{dBW}} \\ &= 10 \cdot \log_{10} \left(\frac{P_1}{1 \text{ mW}} \right) - 10 \cdot \log_{10} \left(\frac{P_2}{1 \text{ mW}} \right) = P_1|_{\text{dBm}} - P_2|_{\text{dBm}} \end{aligned}$$

$\text{dBW} \pm \text{dB}$	$= \text{dBW}$
$\text{dBm} \pm \text{dB}$	$= \text{dBm}$
$\text{dBW} - \text{dBW}$	$= \text{dB}$
$\text{dBm} - \text{dBm}$	$= \text{dB}$

$\text{dBm} + \text{dBm}$
 $\text{dBW} + \text{dBW}$


准则:

$$30\text{dBm}=1\text{ W}$$

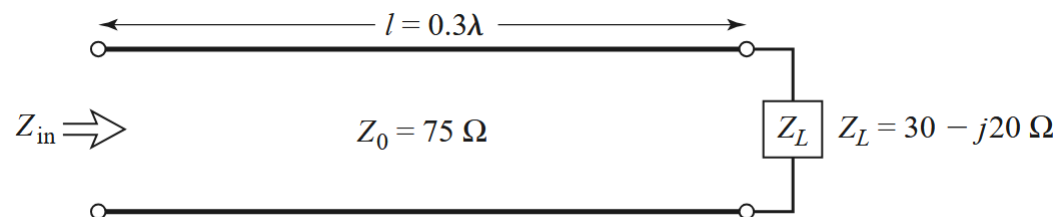
$$X(\text{dBm})+3\text{dB}\rightarrow X*2\text{ mW}$$

$$X(\text{dBm})-3\text{dB}\rightarrow X*1/2\text{ mW}$$

$$X(\text{dBm})+10\text{dB}\rightarrow X*10\text{ mW}$$

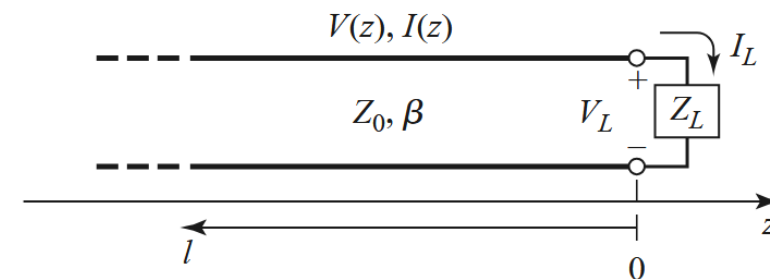
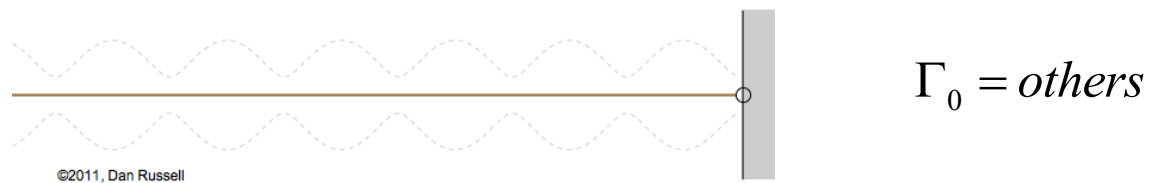
$$X(\text{dBm})-10\text{dB}\rightarrow X*1/10\text{ mW}$$

终端加载的无耗传输线



【例】求传输线上的反射系数分布、输入阻抗 Z_{in} 及电压波分布

终端加载的无耗传输线



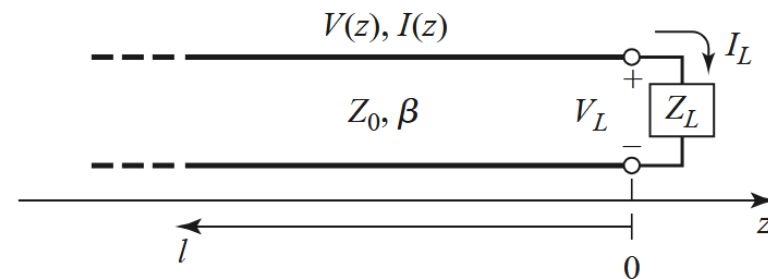
$$U(z) = V_0^+ e^{-j\beta z} (1 + \Gamma_0 e^{2j\beta z})$$

终端加载的无耗传输线

[例] 一根 75Ω 均匀无耗传输线，终端接有负载 $Z_L=R_L+jX_L$ ，欲使线上电压驻波比为3，则负载的实部 R_L 和虚部 X_L 应满足什么关系？

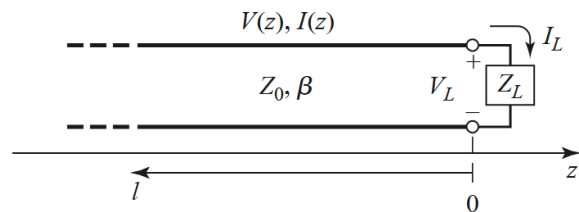
$$|\Gamma_0| = \frac{\rho - 1}{\rho + 1} = 0.5 \quad |\Gamma_0| = \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right| = 0.5$$

$$(R_L - 125)^2 + X_L^2 = 100^2$$



负载的实部 R_L 和虚部 X_L 应在圆心为 $(125, 0)$ 、半径为100的圆上，上半圆对应负载为感抗，而下半圆对应负载为容抗。

终端加载的无耗传输线



$$Z_{in} = \frac{V(-l)}{I(-l)} = \frac{V_o^+ (e^{j\beta l} + \Gamma e^{-j\beta l})}{V_o^+ (e^{j\beta l} - \Gamma e^{-j\beta l})} Z_0 = \frac{1 + \Gamma e^{-2j\beta l}}{1 - \Gamma e^{-2j\beta l}} Z_0$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

具有一定长度的无耗传输线的输入阻抗

$$\begin{aligned}
 Z_{in} &= Z_0 \frac{(Z_L + Z_0)e^{j\beta l} + (Z_L - Z_0)e^{-j\beta l}}{(Z_L + Z_0)e^{j\beta l} - (Z_L - Z_0)e^{-j\beta l}} \\
 &= Z_0 \frac{Z_L \cos \beta l + j Z_0 \sin \beta l}{Z_0 \cos \beta l + j Z_L \sin \beta l} \\
 &= Z_0 \frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l}
 \end{aligned}$$

- $l = \frac{\lambda}{2} \quad Z_{in} = Z_L$

输入阻抗与特征阻抗 Z_0 与传播常数 β 无关

- $l = \frac{\lambda}{4} \quad Z_{in} = Z_0^2 / Z_L$

1. 输入阻抗与与负载阻抗成反比
2. 阻抗变换
3. 短路变开路，开路变短路

- $Z_L = jX_L$

$$Z_{in} = jZ_0 \frac{X_L \cos \beta l + Z_0 \sin \beta l}{Z_0 \cos \beta l - X_L \sin \beta l}$$

纯电抗负载导致输入阻抗也为纯电抗

- $l \ll \lambda, \beta l \approx 0$

$Z_{in} = Z_L$ ，即短线近似

负载短路

$$V(z) = V_o^+ (e^{-j\beta z} - e^{j\beta z}) = -2j V_o^+ \sin \beta z,$$

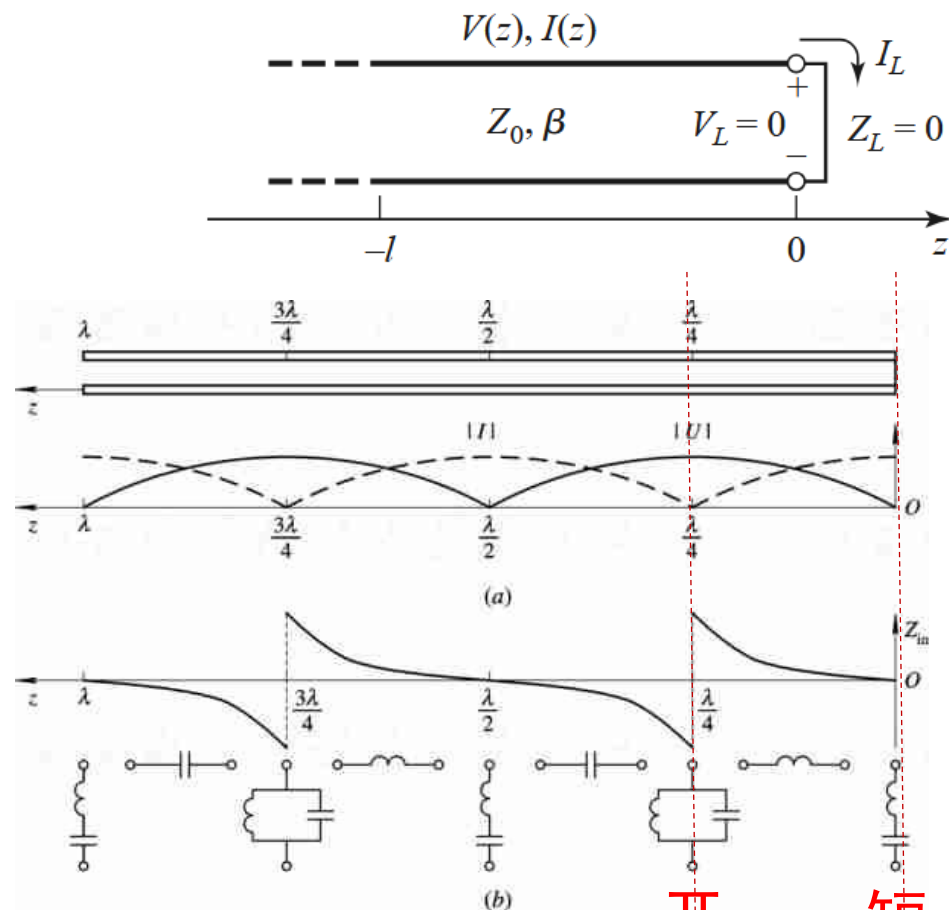
$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} + e^{j\beta z}) = \frac{2V_o^+}{Z_0} \cos \beta z,$$

$$Z_{in} = j Z_0 \tan \beta \ell,$$

$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\} = 2|V_o^+| \sin \beta z \sin(\omega t + \varphi_0)$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\} = \frac{2|V_o^+|}{Z_0} \cos \beta z \cos(\omega t + \varphi_0)$$

v/i 相位差了180度



开路点

短路点

负载开路

$$V(z) = V_o^+ (e^{-j\beta z} + e^{j\beta z}) = 2V_o^+ \cos \beta z,$$

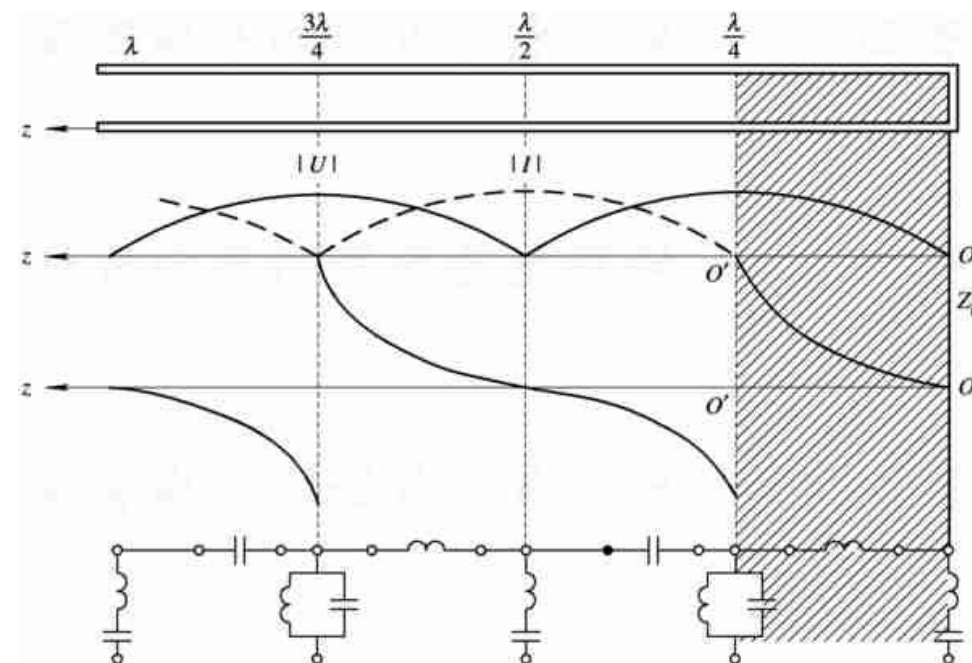
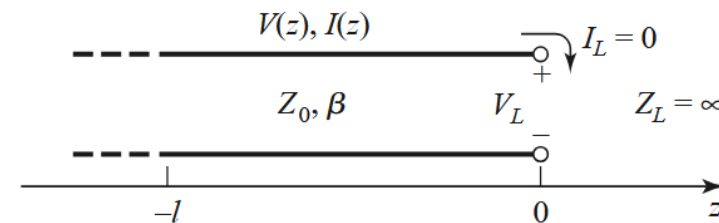
$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - e^{j\beta z}) = \frac{-2jV_o^+}{Z_0} \sin \beta z,$$

$$Z_{in} = -jZ_0 \cot \beta \ell,$$

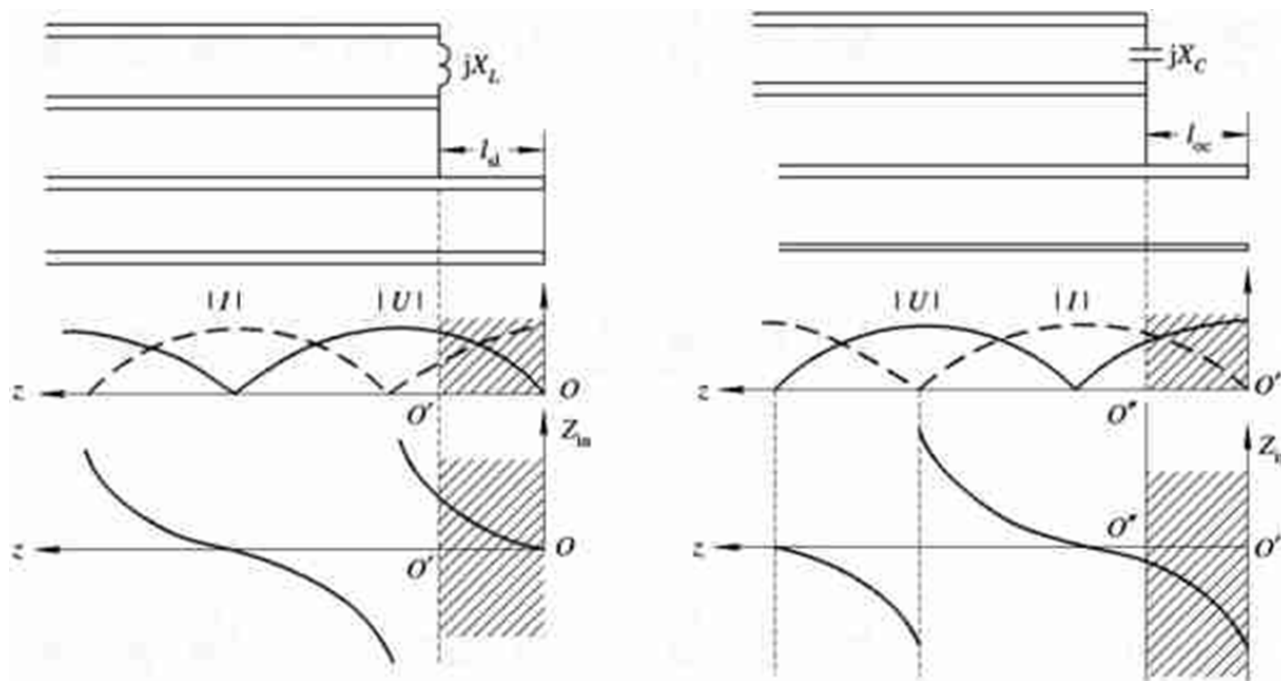
$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\} = 2|V_o^+| \cos \beta z \cos(\omega t + \varphi_0)$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\} = \frac{2|V_o^+|}{Z_0} \sin \beta z \sin(\omega t + \varphi_0)$$

v/i 相位差了180度

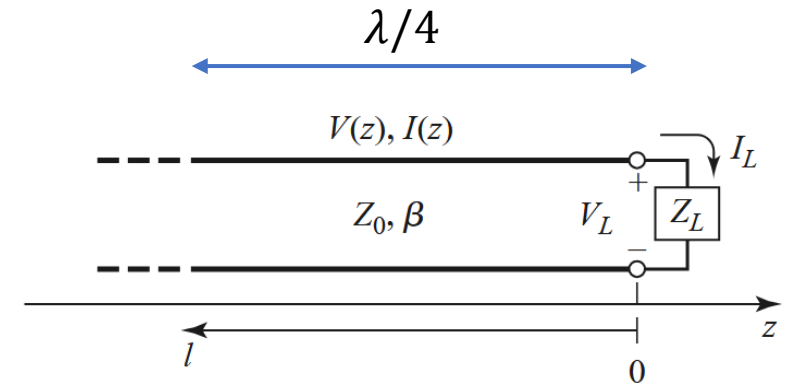


开路/短路状态下的负载特性分析

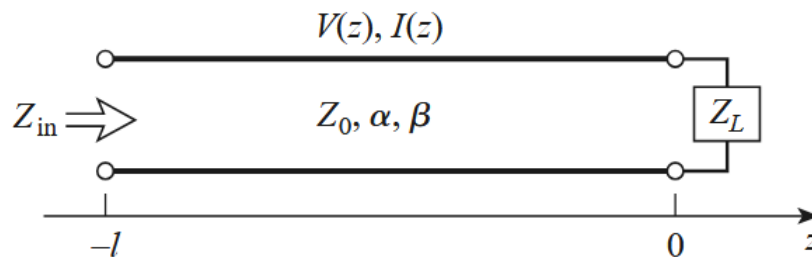


$\lambda/4$ 阻抗变换

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} = \frac{Z_0^2}{Z_L}$$



有耗传输线



损耗因子

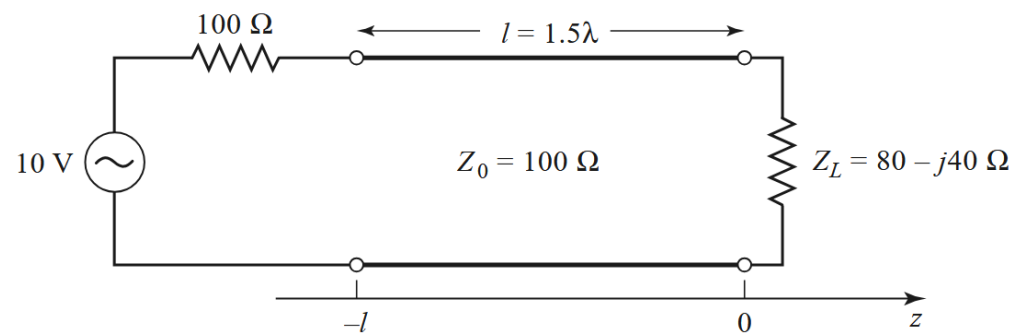
$$\gamma = \alpha + j\beta$$

$$\begin{aligned}
 V(z) &= V_o^+ (e^{-\gamma z} + \Gamma e^{\gamma z}), \\
 I(z) &= \frac{V_o^+}{Z_0} (e^{-\gamma z} - \Gamma e^{\gamma z}), \\
 Z_{\text{in}} &= \frac{V(-\ell)}{I(-\ell)} = Z_0 \frac{Z_L + Z_0 \tanh \gamma \ell}{Z_0 + Z_L \tanh \gamma \ell}, \quad P_L = \frac{1}{2} \text{Re}\{V(0)I^*(0)\} = \frac{|V_o^+|^2}{2Z_0} (1 - |\Gamma|^2), \\
 P_{\text{in}} &= \frac{1}{2} \text{Re}\{V(-\ell)I^*(-\ell)\} = \frac{|V_o^+|^2}{2Z_0} (e^{2\alpha\ell} - |\Gamma|^2 e^{-2\alpha\ell}) \\
 &= \frac{|V_o^+|^2}{2Z_0} (1 - |\Gamma(\ell)|^2) e^{2\alpha\ell}, \quad P_{\text{loss}} = P_{\text{in}} - P_L = \frac{|V_o^+|^2}{2Z_0} [(e^{2\alpha\ell} - 1) + |\Gamma|^2 (1 - e^{-2\alpha\ell})].
 \end{aligned}$$

入射波损耗

反射波损耗

终端加载的无耗传输线



【例】求传输线上的电压波幅值分布

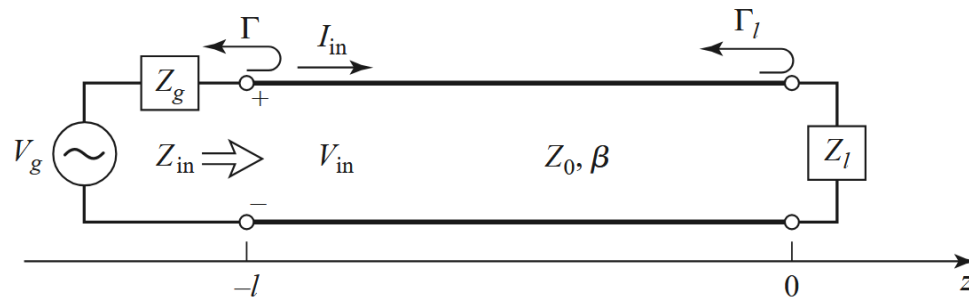
源与负载匹配

$$Z_{\text{in}} = Z_0 \frac{1 + \Gamma_{\ell} e^{-2j\beta\ell}}{1 - \Gamma_{\ell} e^{-2j\beta\ell}} = Z_0 \frac{Z_{\ell} + jZ_0 \tan \beta\ell}{Z_0 + jZ_{\ell} \tan \beta\ell},$$

$$\Gamma_{\ell} = \frac{Z_{\ell} - Z_0}{Z_{\ell} + Z_0}, \quad V(z) = V_o^+ (e^{-j\beta z} + \Gamma_{\ell} e^{j\beta z}),$$

$$V(-\ell) = V_g \frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g} = V_o^+ (e^{j\beta\ell} + \Gamma_{\ell} e^{-j\beta\ell}),$$

$$V_o^+ = V_g \frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g} \frac{1}{(e^{j\beta\ell} + \Gamma_{\ell} e^{-j\beta\ell})} \Rightarrow V_o^+ = V_g \frac{Z_0}{Z_0 + Z_g} \frac{e^{-j\beta\ell}}{(1 - \Gamma_{\ell} \Gamma_g e^{-2j\beta\ell})}, \quad \Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}.$$



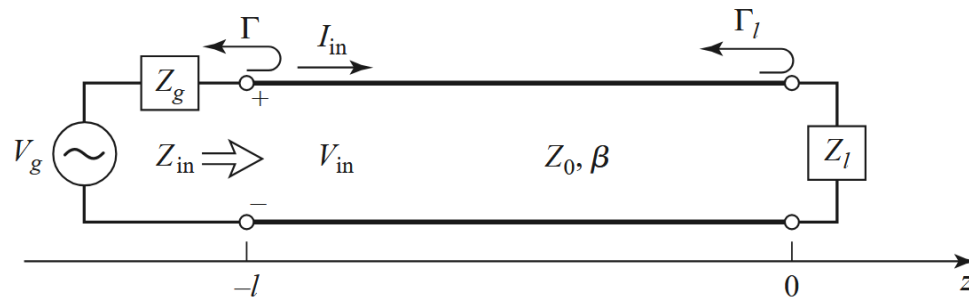
$$P = \frac{1}{2} \text{Re}\{V_{\text{in}} I_{\text{in}}^*\} = \frac{1}{2} |V_{\text{in}}|^2 \text{Re}\left\{\frac{1}{Z_{\text{in}}}\right\} = \frac{1}{2} |V_g|^2 \left|\frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g}\right|^2 \text{Re}\left\{\frac{1}{Z_{\text{in}}}\right\}.$$

$$P = \frac{1}{2} |V_g|^2 \frac{R_{\text{in}}}{(R_{\text{in}} + R_g)^2 + (X_{\text{in}} + X_g)^2}.$$

源与负载匹配

负载与传输线匹配 $Z_L = Z_0$

$$P = \frac{1}{2} |V_g|^2 \frac{Z_0}{(Z_0 + R_g)^2 + X_g^2}$$



源与传输线匹配 $Z_{in} = Z_g$

$$P = \frac{1}{2} |V_g|^2 \frac{R_g}{4(R_g^2 + X_g^2)}$$

$$P = \frac{1}{2} |V_g|^2 \frac{R_{in}}{(R_{in} + R_g)^2 + (X_{in} + X_g)^2}$$

$$\frac{\partial P}{\partial R_{in}} = 0 \rightarrow \frac{1}{(R_{in} + R_g)^2 + (X_{in} + X_g)^2} + \frac{-2R_{in}(R_{in} + R_g)}{[(R_{in} + R_g)^2 + (X_{in} + X_g)^2]^2} = 0, \quad \Rightarrow \quad Z_{in} = Z_g^*$$

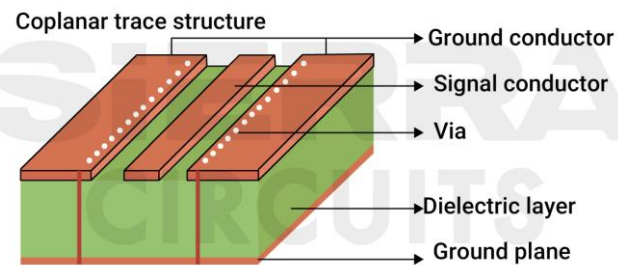
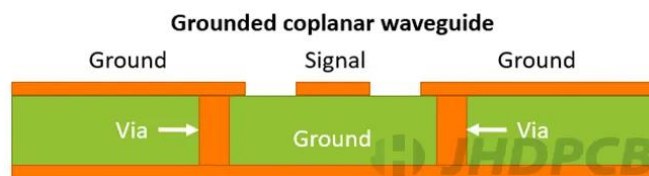
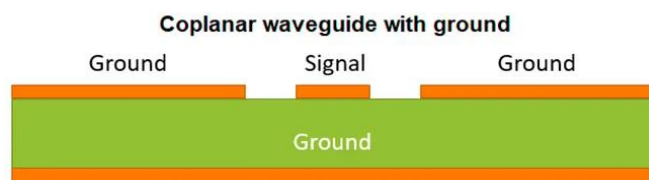
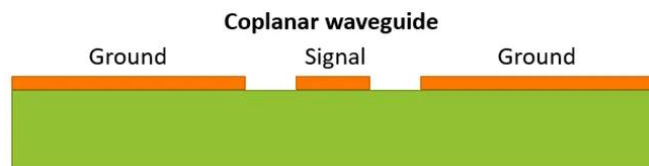
共轭匹配

$$P = \frac{1}{2} |V_g|^2 \frac{1}{4R_g}, \quad \text{但是效率仅有50\%, 若提高效率需尽量降低} Z_g$$

两种情况下的功率谁大谁小?

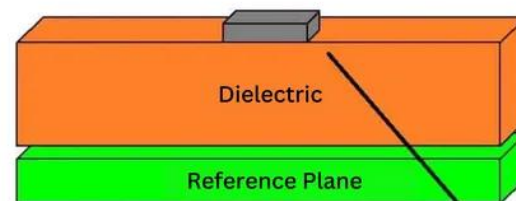
平面传输线

共面波导 Coplanar waveguide, CPW

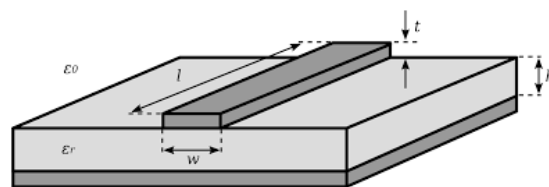


微带传输线 Microstrip line

Microstrip

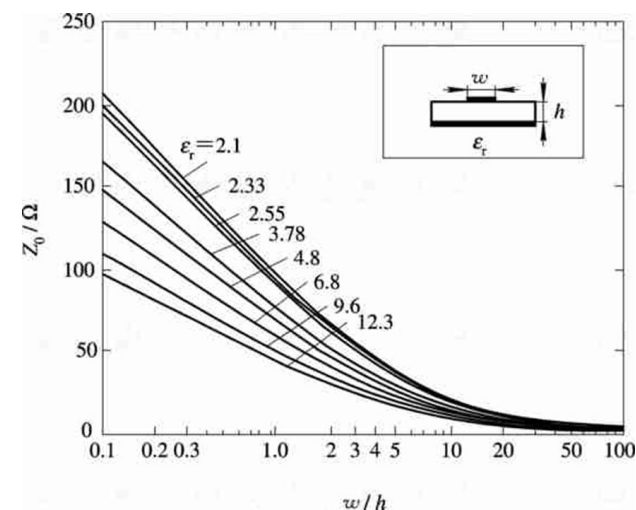
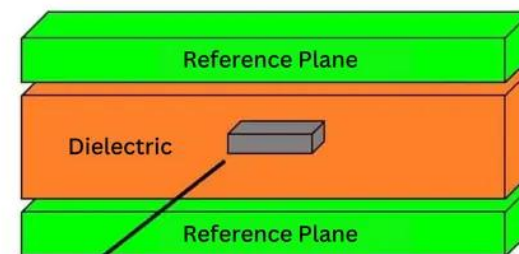


High-speed transmission line



带状线

Stripline



微波电路原理与设计

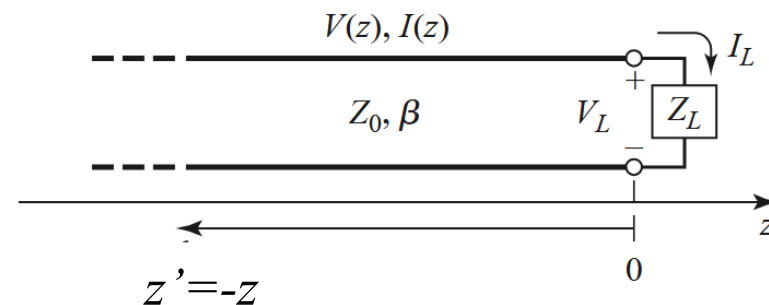
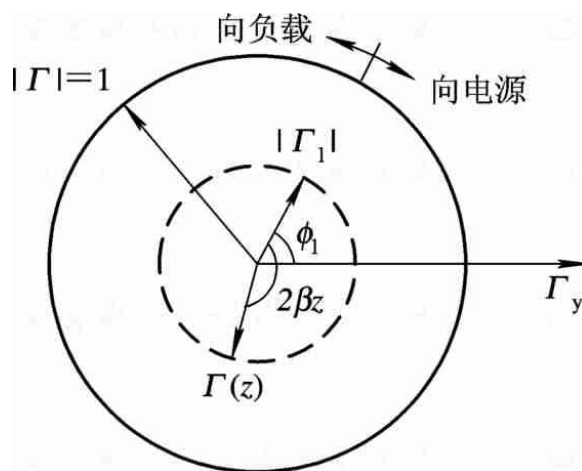
第一章 史密斯圆图及阻抗匹配

张允晶
2026-03

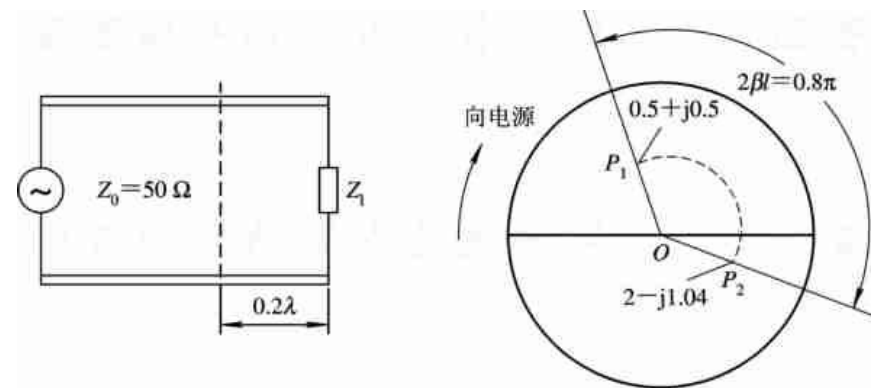
史密斯阻抗圆图

$$\Gamma(z) = \frac{Z_{in}(z') - Z_0}{Z_{in}(z') + Z_0} = \frac{\bar{Z}_{in}(z') - 1}{\bar{Z}_{in}(z') + 1}$$

$$\Gamma(z') = \frac{V_0^- e^{-j\beta z'}}{V_0^+ e^{j\beta z'}} = \Gamma_0 e^{-j2\beta z'}$$



[例] 已知传输线的特性阻抗 $Z_0=50\Omega$ ，如图1-22所示。假设传输线的负载阻抗为 $Z_1=25+j25\Omega$ ，求离负载 $z=0.2\lambda$ 处的等效阻抗。

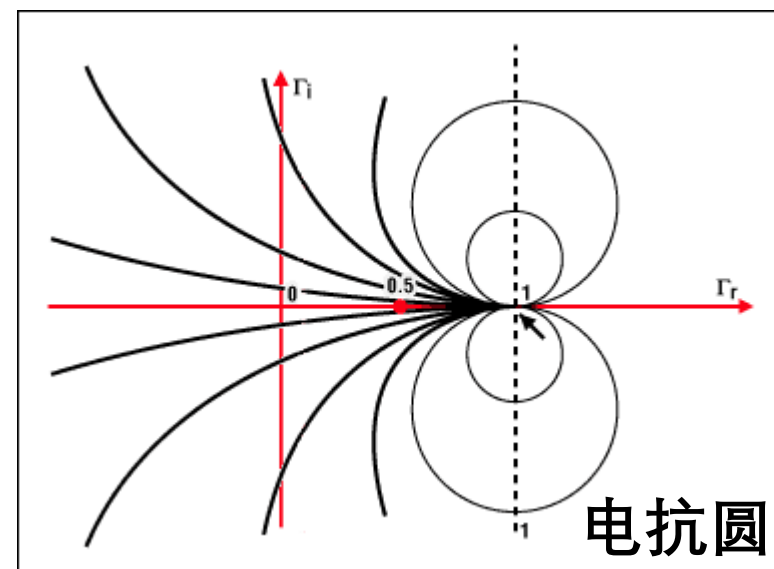
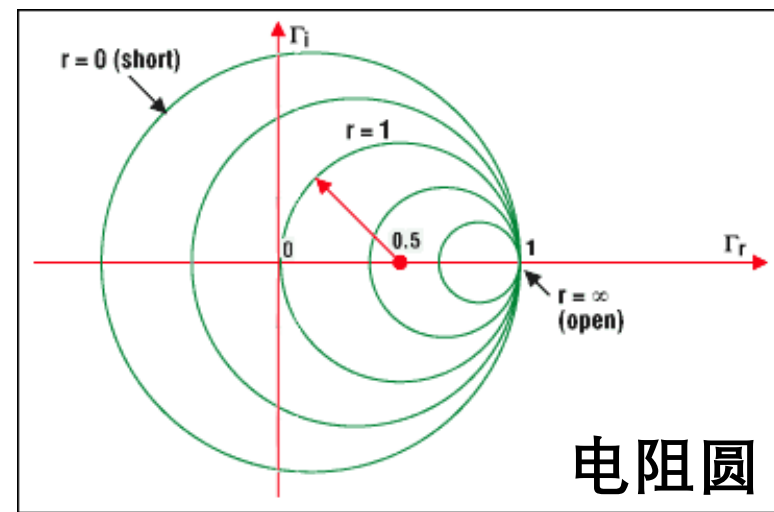
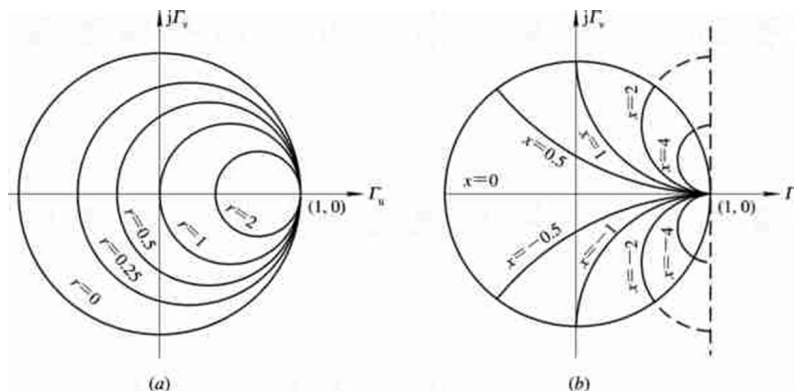


史密斯阻抗圆图

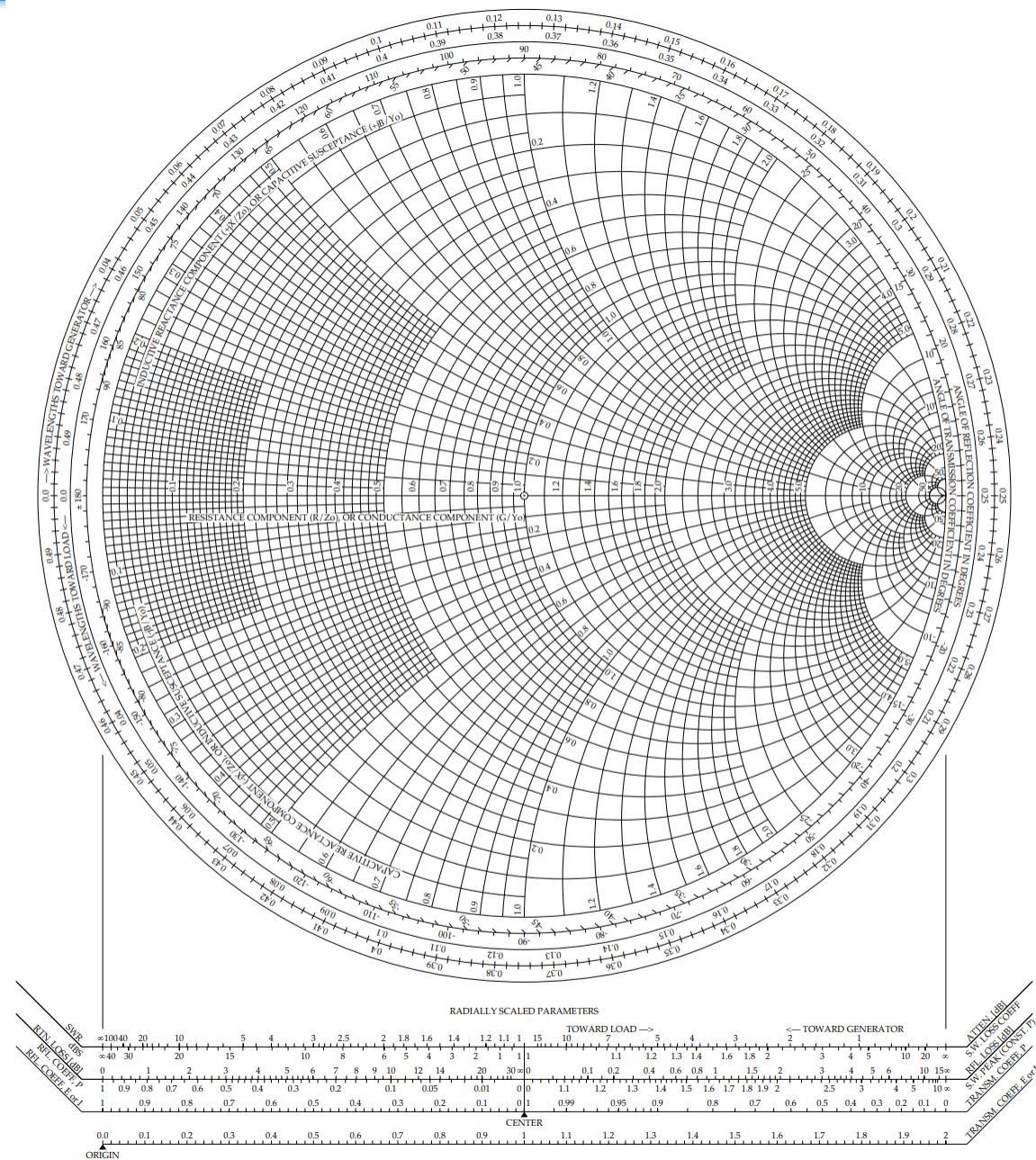
$$\Gamma(z) = \Gamma_r + j\Gamma_i \quad \bar{Z}_{in} = \frac{Z_{in}}{Z_0} = \frac{1 + (\Gamma_r + j\Gamma_i)}{1 - (\Gamma_r + j\Gamma_i)}$$

令 $\bar{Z}_{in} = r + jx$, 则可得以下方程:

$$\left\{ \begin{aligned} \left(\Gamma_r - \frac{r}{1+r} \right)^2 + \Gamma_i^2 &= \left(\frac{1}{1+r} \right)^2 \\ (\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x} \right)^2 &= \left(\frac{1}{x} \right)^2 \end{aligned} \right.$$



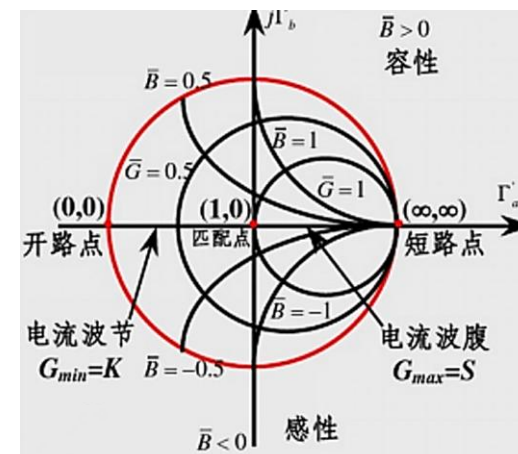
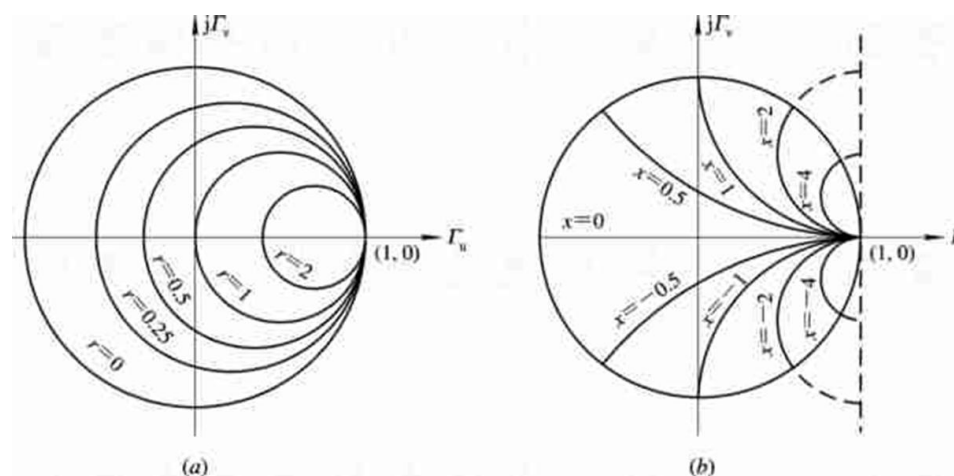
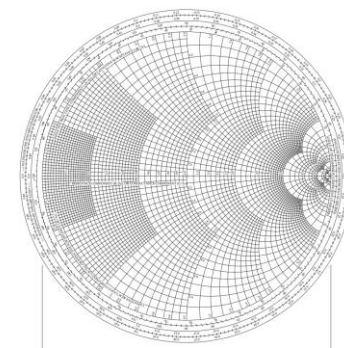
史密斯阻抗圆图



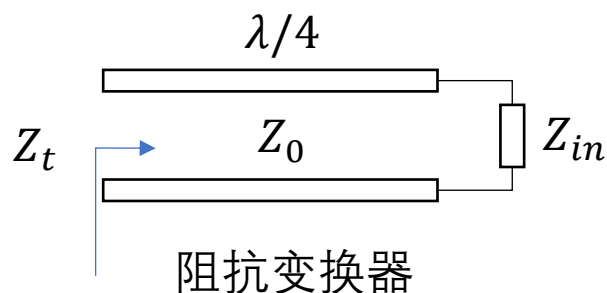
史密斯阻抗圆图

$$\rho = \text{SWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

1. 阻抗圆图上半圆内的电抗 $x > 0$ 呈感性，下半圆内的电抗 $x < 0$ 呈容性。
2. 实轴上的点代表纯电阻点，左半轴上的点为电压波节点，代表行波系数 k (Why?)
3. 右半轴上的点为电压波腹点，其上的刻度代表驻波比 ρ (Why?)
4. 圆图旋转一周为 $\lambda/2$
5. $|\Gamma| = 1$ 的圆周上的点代表纯电抗点
6. 实轴左端点为短路点，右端点为开路点，中心点是匹配点
7. 在传输线上由负载向电源方向移动时，在圆图上应顺时针旋转；反之，由电源向负载方向移动时，应逆时针旋转。

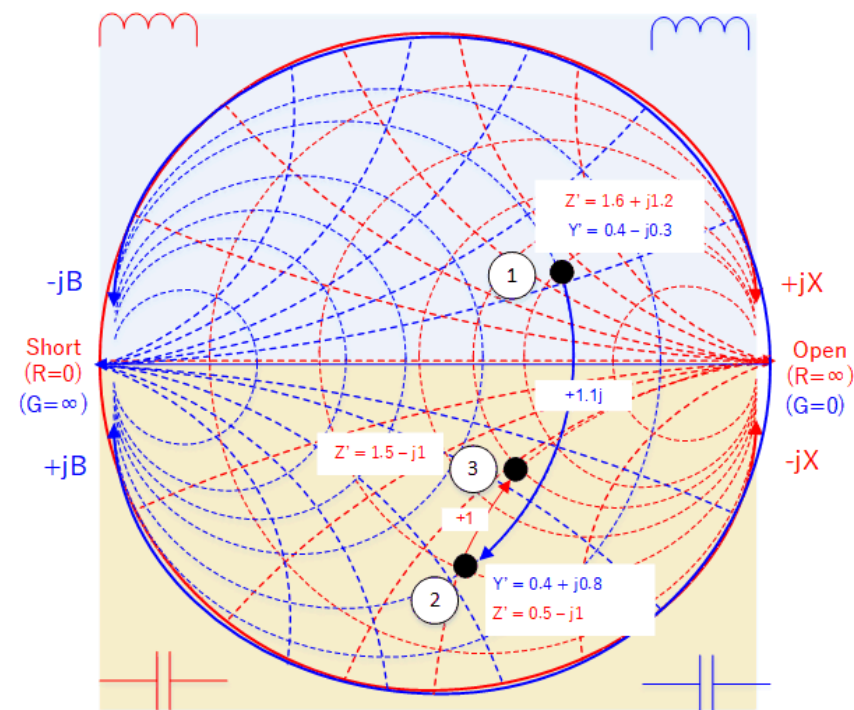
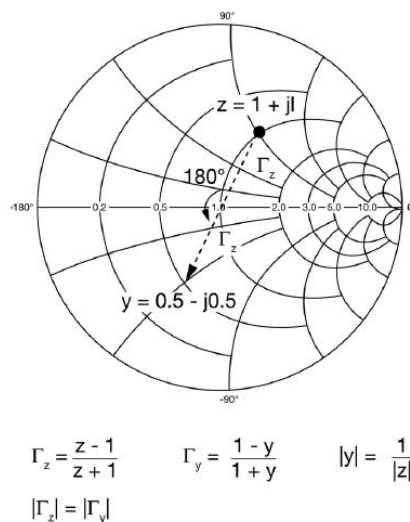


史密斯导纳圆图



$$Z_t = Z_0^2 / Z_{in}$$

$$z_t = \frac{Z_t}{Z_0} = \frac{Z_0}{Z_{in}} = \frac{1}{z_{in}} = y_{in} = g + jb$$



- ◆ 经过1/4波长，原来的归一化阻抗变成了归一化导纳，
在阻抗圆图直接读出的数值即为转化后的导纳值
- ◆ Z转化为Y后，其反射系数的幅值不变，相位发生180度变化

史密斯圆图的应用-串联RL

$$z_{in}(\omega_L) = r + j \frac{\omega_L L}{Z_0}$$

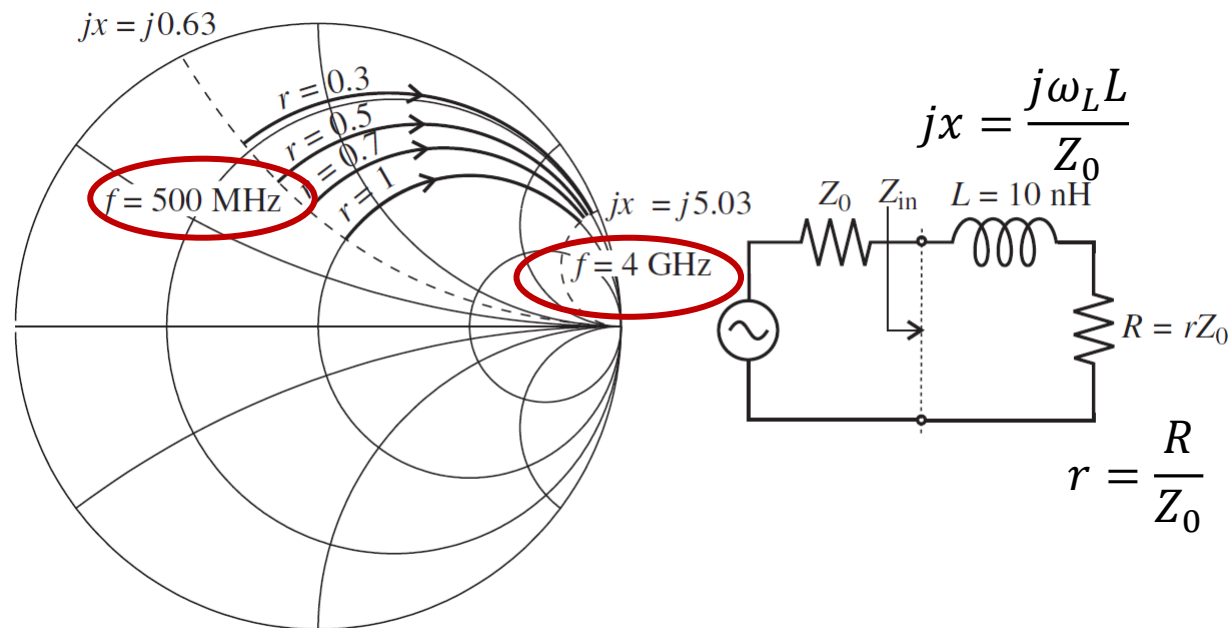
- 在等电阻圆上走，相当于串联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 L)
- 在等电抗圆上走，相当于串联电抗不变，增加或减少一定量的电阻 (变化的为 r)

$$jx(500 \text{ MHz}) = j \frac{2\pi \times 500e6 \times 10e-9}{50} = j0.63$$

增加频率 ↓ 等电阻圆上从j0.63→j5.03

$$jx(4000 \text{ MHz}) = j \frac{2\pi \times 4000e6 \times 10e-9}{50} = j5.03$$

阻抗图



史密斯圆图的应用-串联RL

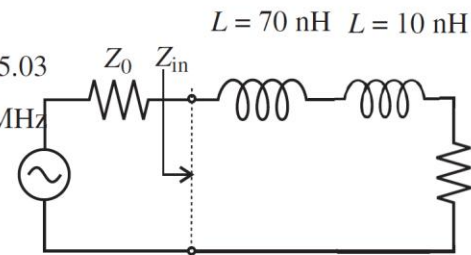
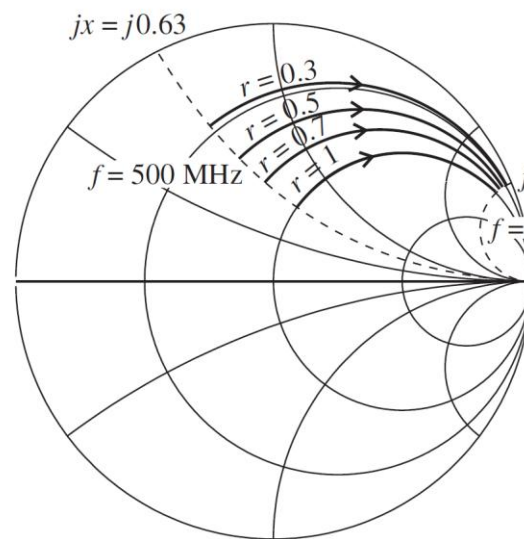
串联（增加）电感

$$z_{in}(\omega_L) = r + jx + j \frac{\omega_L \Delta L}{Z_0}$$

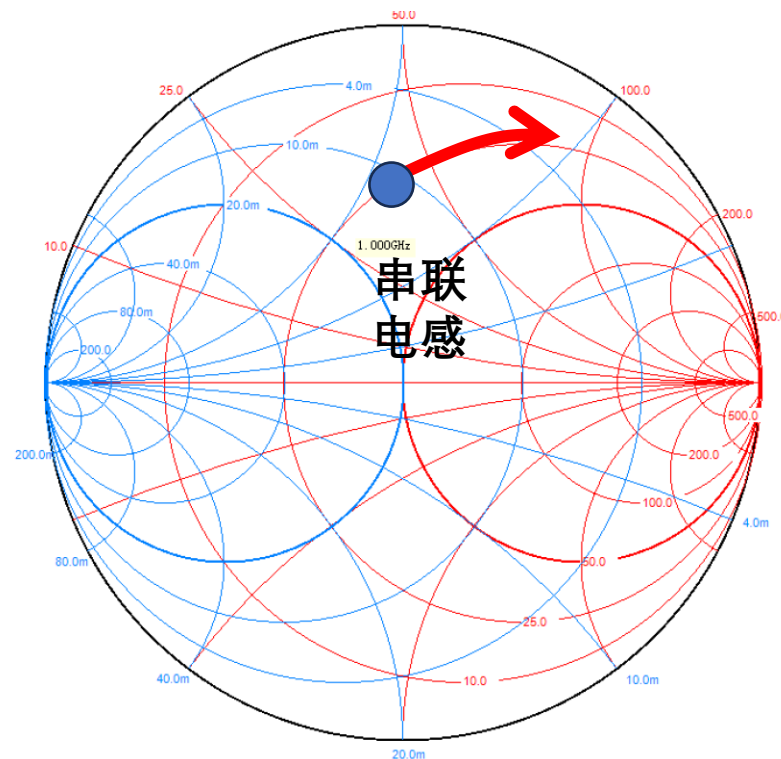
$$jx(10 \text{ nH}) = j \frac{2\pi \times 500e6 \times 10e-9}{50} = j0.63$$

↓ 等电阻圆上从j0.63→j5.03

$$\begin{aligned} jx(10 \text{ nH} + 70 \text{ nH}) &= j0.63 + j \frac{2\pi \times 500e6 \times 70e-9}{50} \\ &= j0.63 + j4.4 = j5.03 \end{aligned}$$



串联电感，沿电阻圆顺时针走



史密斯圆图的应用-串联RC

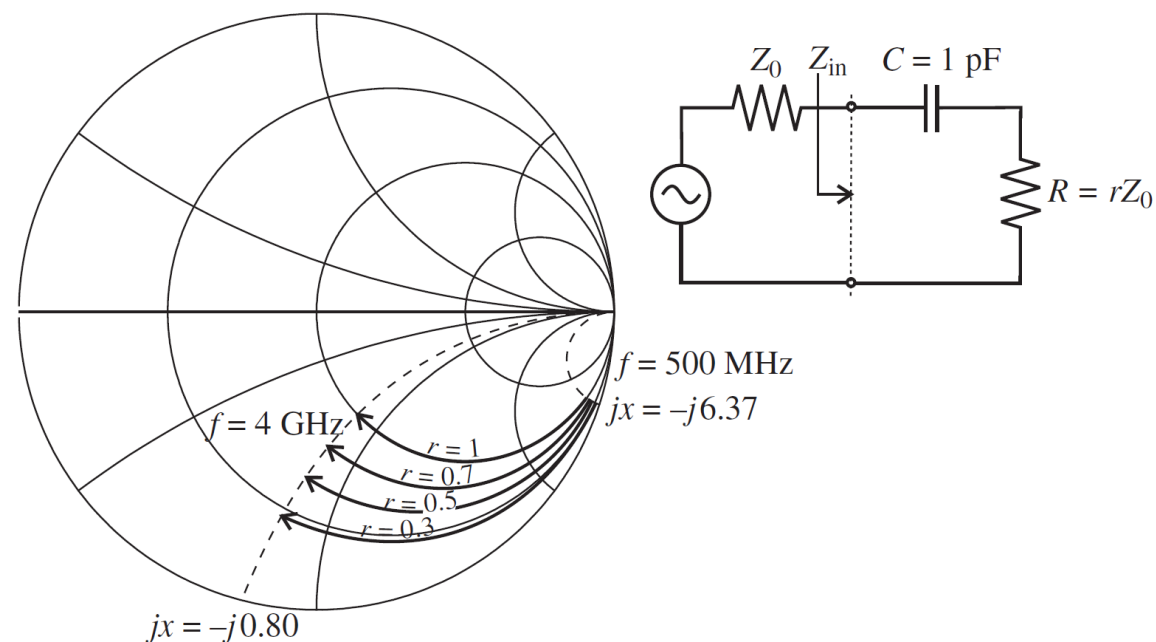
$$z_{in}(\omega_L) = r - j \frac{1}{\omega C Z_0}$$

- 在等电阻圆上走，相当于串联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 C)
- 在等电抗圆上走，相当于串联电抗不变，增加或减少一定量的电阻 (变化的为 r)

$$jx(500 \text{ MHz}) = -j \frac{1}{\omega_0 C Z_0} = -j6.37$$

增加频率 ↓

$$jx(4000 \text{ MHz}) = -j \frac{1}{\omega_1 C Z_0} = -j0.8$$

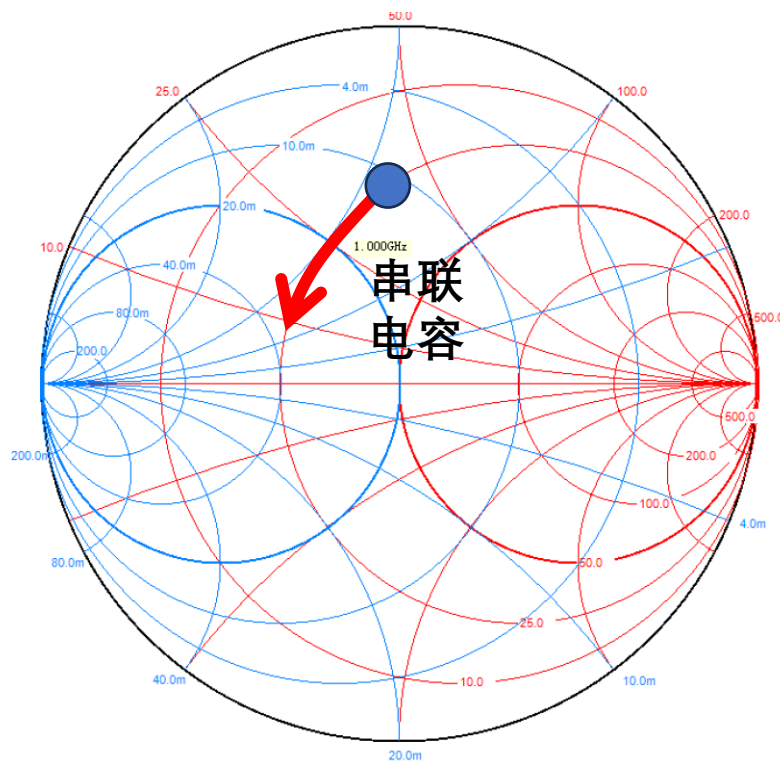


史密斯圆图的应用-串联RC

串联（增加）电容

$$z_{in}(\omega_L) = r + jx - j \frac{1}{\omega_L \Delta C Z_0}$$

串联电容，沿电阻圆逆时针走

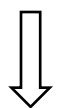


史密斯圆图的应用-并联RL

$$y_{in}(\omega_L) = g - j \frac{Z_0}{\omega_L L}$$

- 在**等电导圆**上走，相当于并联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 L)
- 在**等电纳圆**上走，相当于并联电抗不变，增加或减少一定量的电导（变化的为 g ）

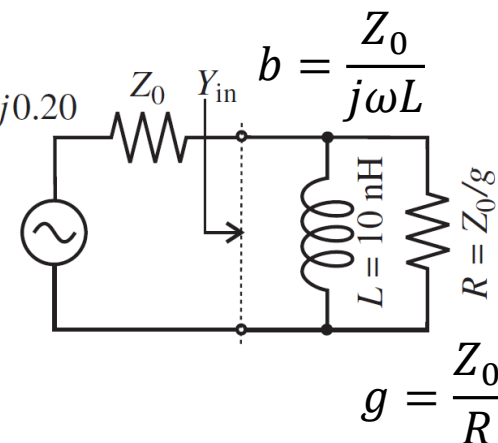
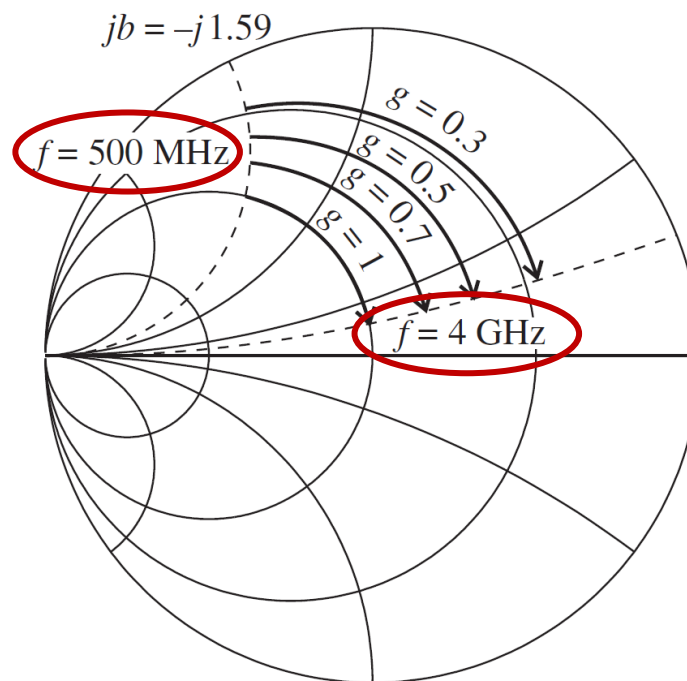
$$jb(500 \text{ MHz}) = -\frac{j50}{2\pi \times 500e6 \times 10e-9} = -j1.59$$



等电导圆上从-j1.59→-j0.2

$$jb(4000 \text{ MHz}) = -\frac{j50}{2\pi \times 4000e6 \times 10e-9} = -j0.2$$

导纳图

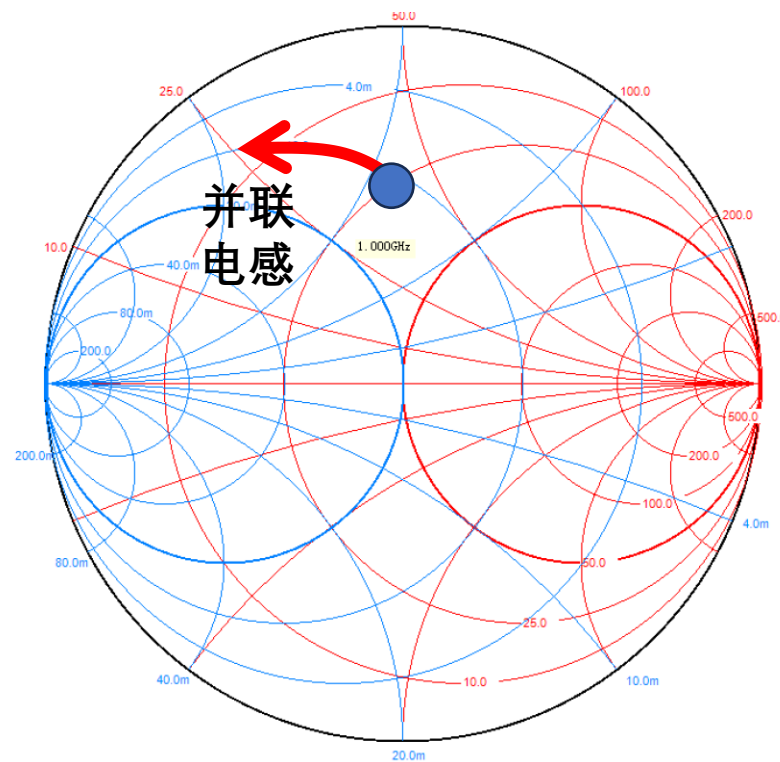


史密斯圆图的应用-并联RL

并联电感

$$z_{in}(\omega_L) = g + jb - j \frac{Z_0}{\omega_L \Delta L}$$

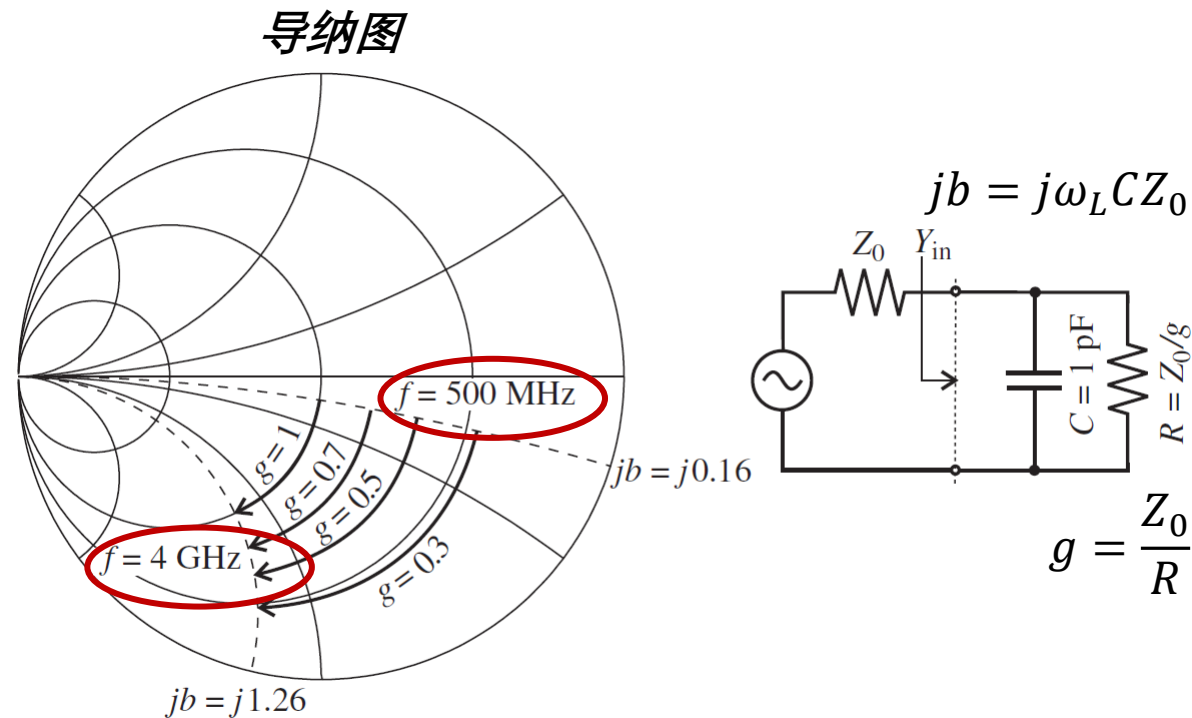
并联电感，沿电导圆逆时针走



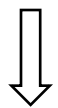
史密斯圆图的应用-并联RC

$$y_{in}(\omega_L) = g + j\omega_L C Z_0$$

- 在**等电导圆**上走，相当于并联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 C)
- 在**等电纳圆**上走，相当于并联电抗不变，增加或减少一定量的电导(变化的为 g)



$$jb(500 \text{ MHz}) = j2\pi \times 500e6 \times 50 \times 1e-12 = j0.16$$



等电导圆上从j0.16→j1.26

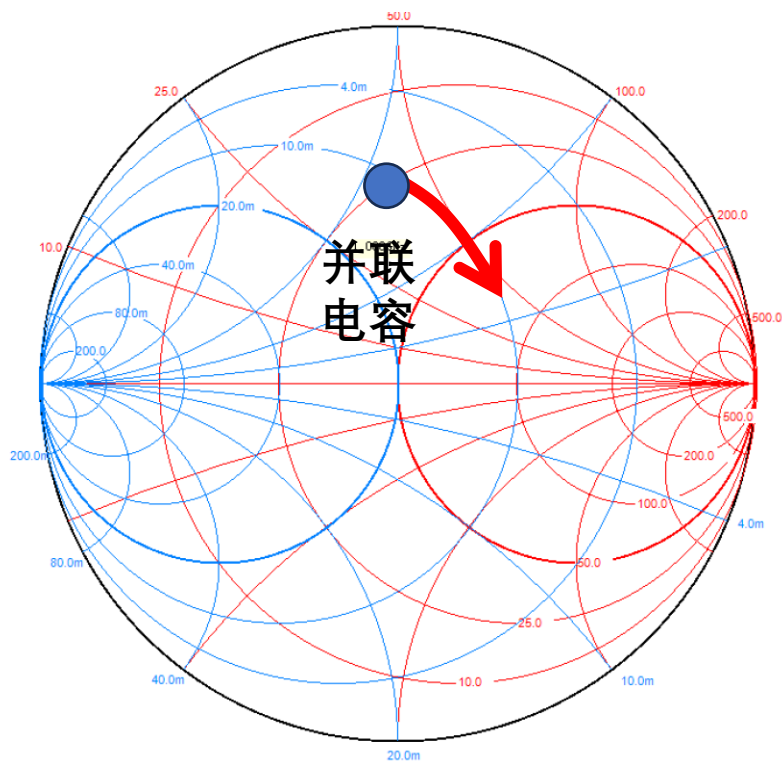
$$jb(4000 \text{ MHz}) = j2\pi \times 4000e6 \times 50 \times 1e-12 = j1.26$$

史密斯圆图的应用-并联RL

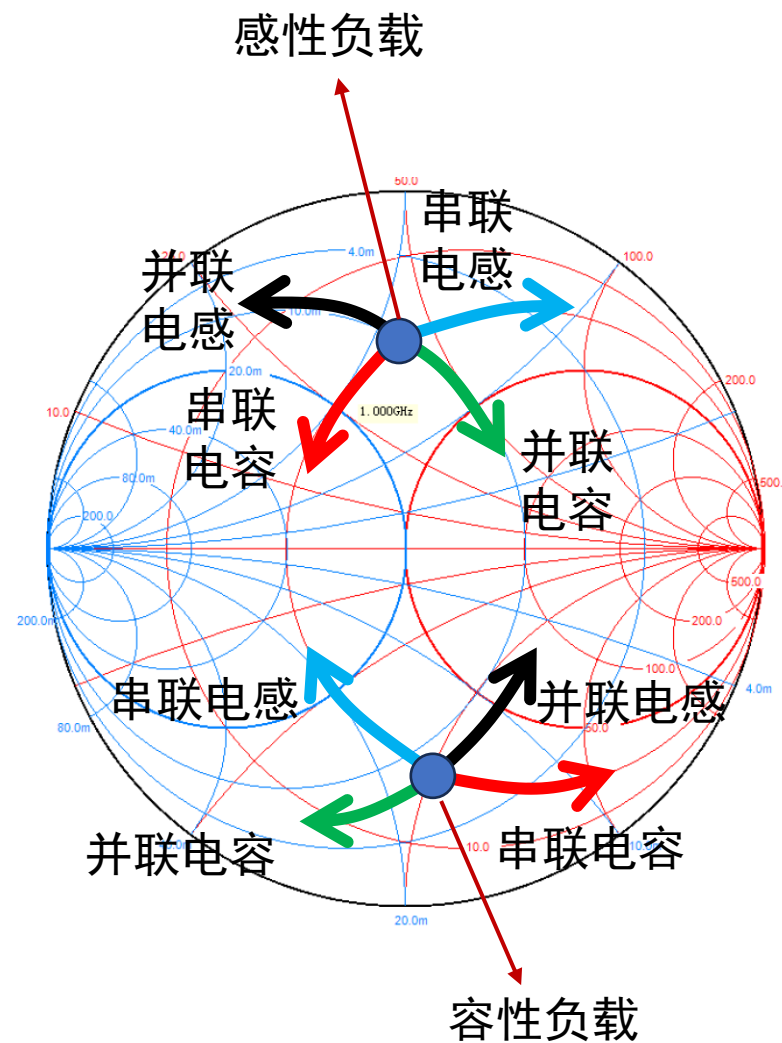
并联电容

$$z_{in}(\omega_L) = g + jb + j\omega\Delta CZ_0$$

并联电容，沿电导圆顺时针走

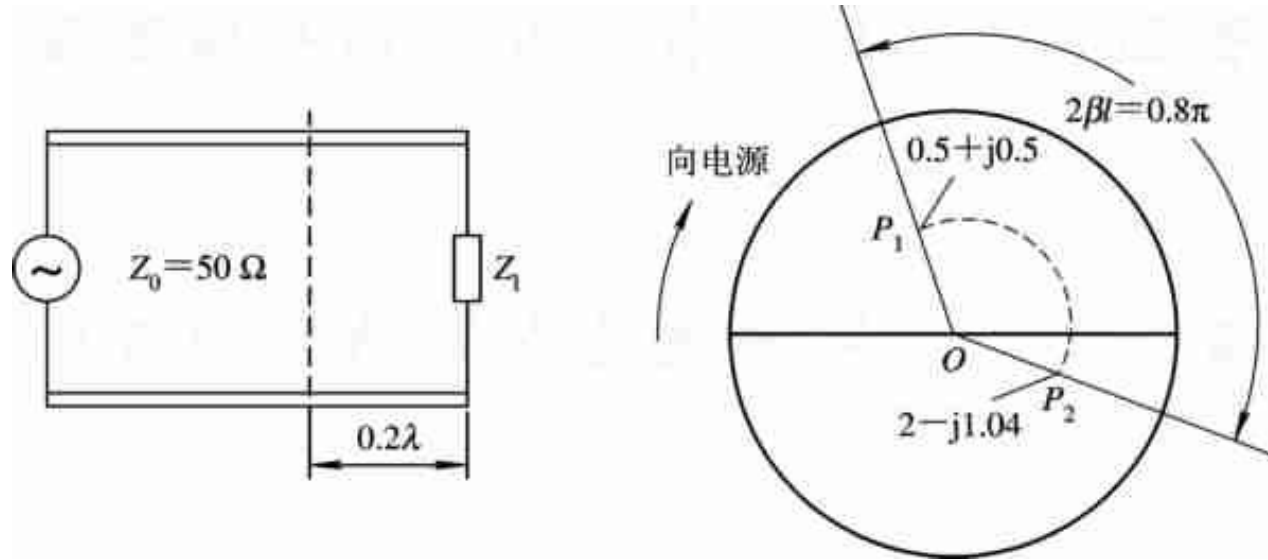


史密斯圆图的应用-串并联电感/电容



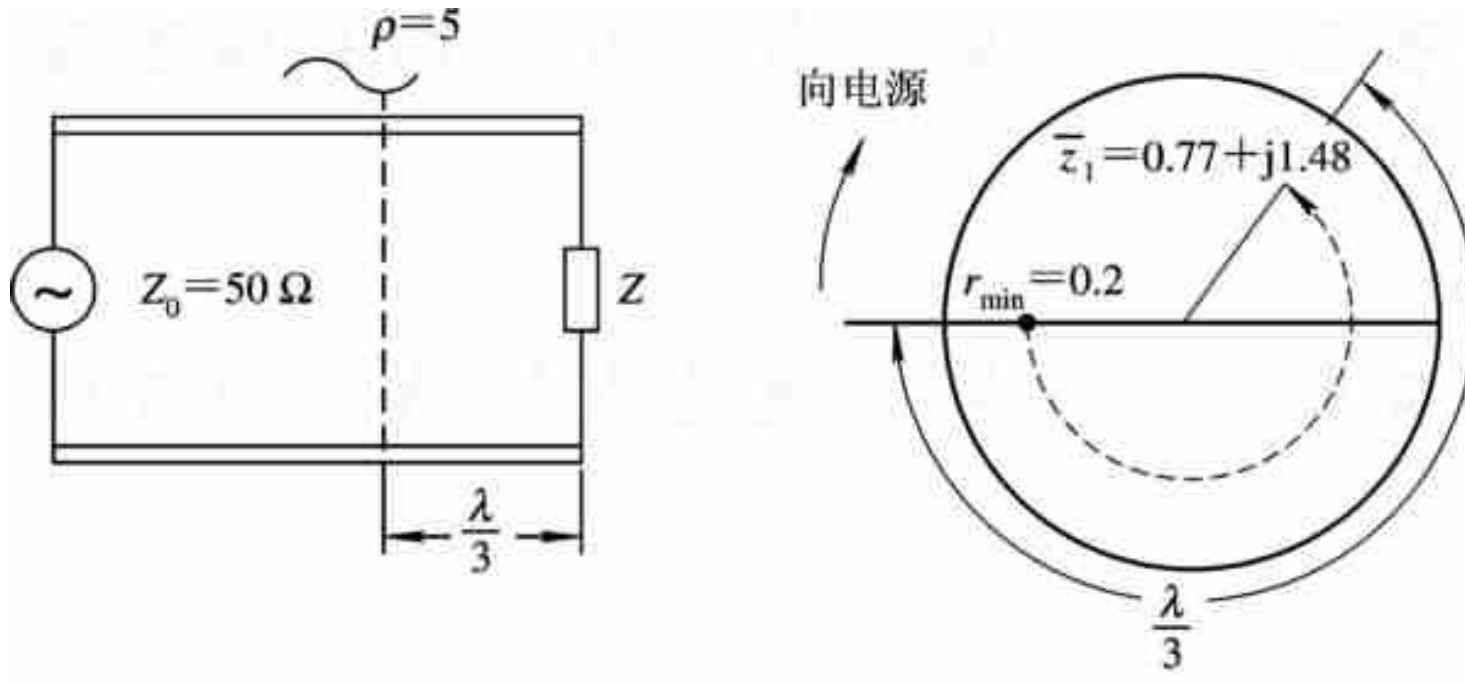
史密斯圆图的应用-传输线的影响

[例] 已知传输线的特性阻抗 $Z_0=50\Omega$ ，如图1-22所示。假设传输线的负载阻抗为 $Z_L=25+j25\Omega$ ，求离负载 $z=0.2\lambda$ 处的等效阻抗。

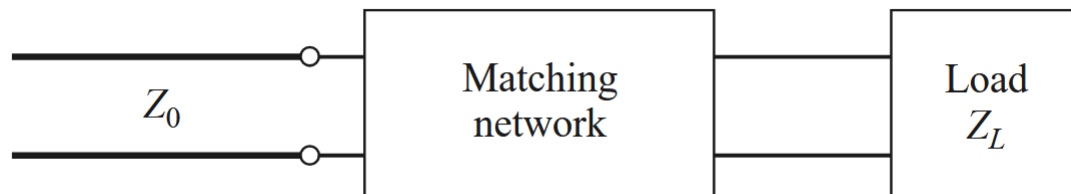


史密斯圆图的应用-传输线的影响

〔例〕 在特性阻抗 $Z_0=50\Omega$ 的无耗传输线上测得驻波比 $\rho=5$ ，电压最小点出现在 $z=\lambda/3$ 处，求负载阻抗。

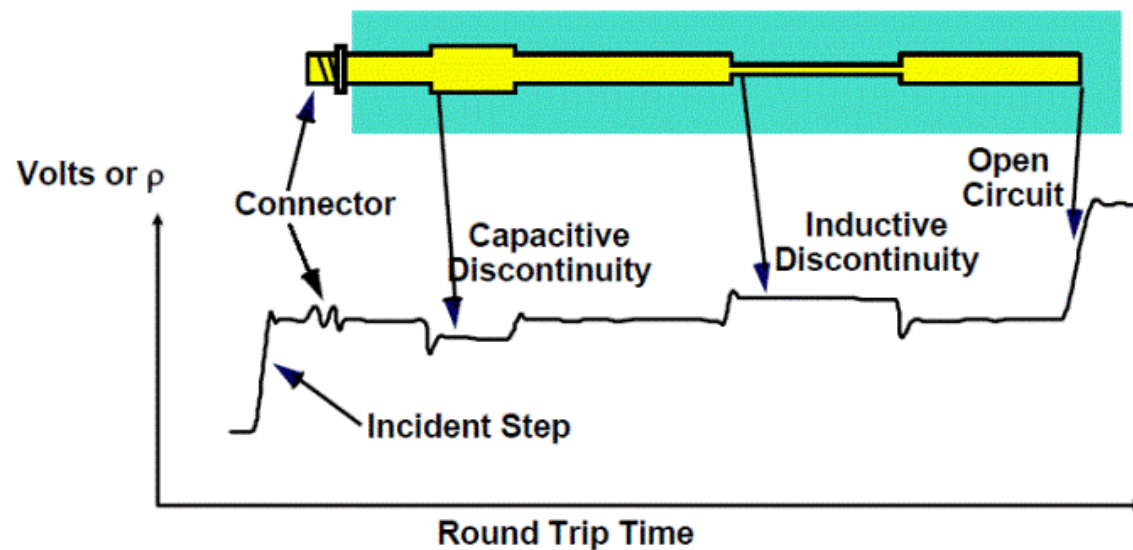


阻抗匹配

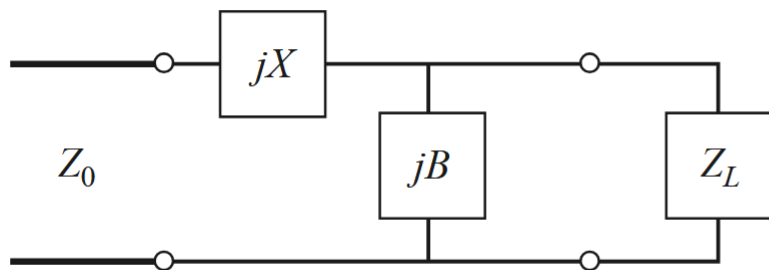


阻抗匹配考虑原则:

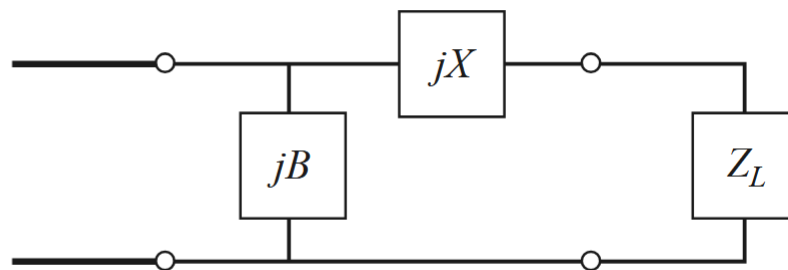
1. 一般情况下，要求匹配网络无损耗元件，且整体损耗低
2. 结构是否简单？
3. 带宽如何？
4. 实施难度如何？
5. 是否可调？



集总元件阻抗匹配



(a)



(b)

$$Z_0 = jX + \frac{1}{jB + 1/(R_L + jX_L)} \Rightarrow$$

$$\begin{aligned} B(XR_L - X_L Z_0) &= R_L - Z_0, \\ X(1 - BX_L) &= BZ_0 R_L - X_L. \end{aligned}$$

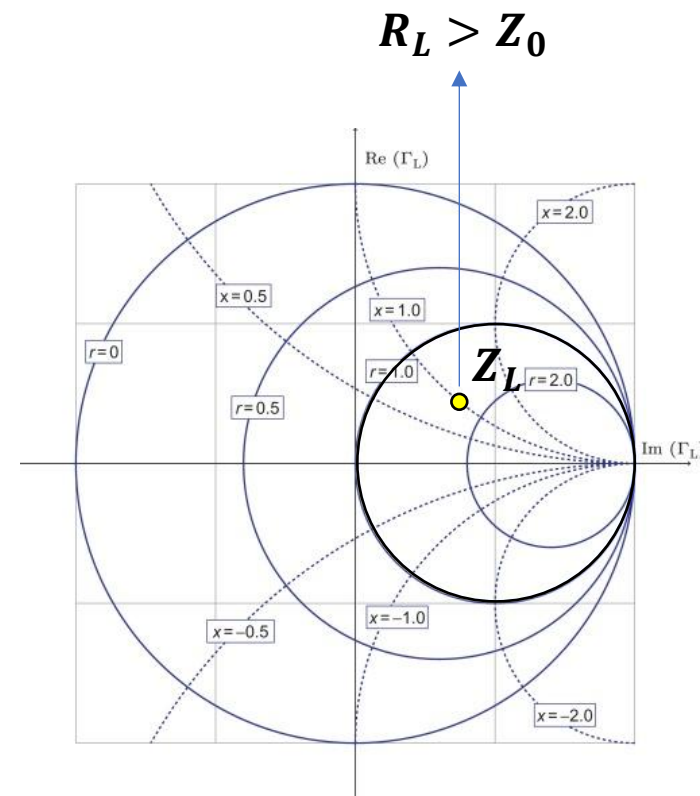
$$\Rightarrow B = \frac{X_L \pm \sqrt{R_L/Z_0} \sqrt{R_L^2 + X_L^2 - Z_0 R_L}}{R_L^2 + X_L^2}.$$

$$\xRightarrow{R_L > Z_0} X = \frac{1}{B} + \frac{X_L Z_0}{R_L} - \frac{Z_0}{B R_L} \xRightarrow{\quad}$$

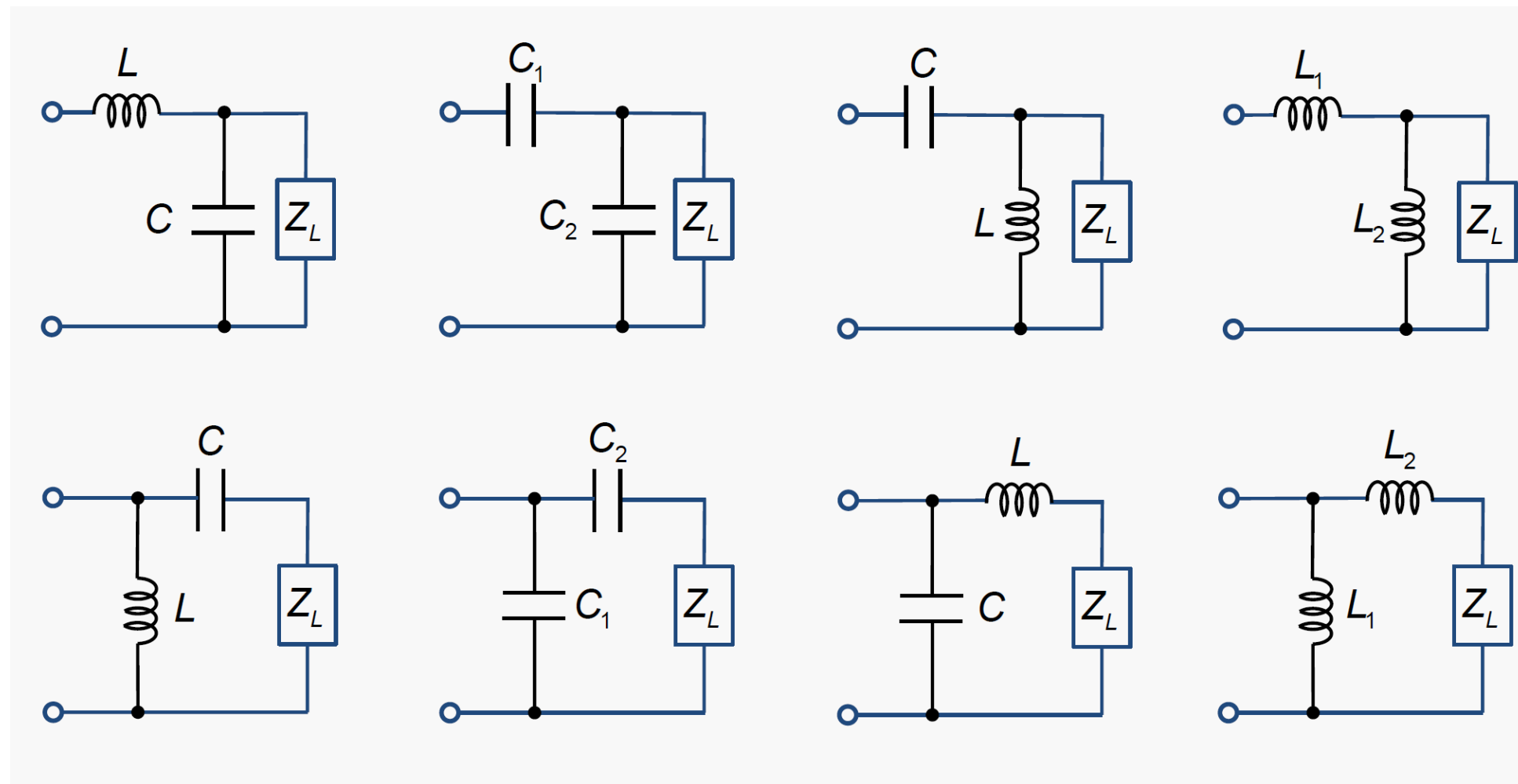
存在两种匹配方式:

B可以为正或为负 (电容或者电感)

X可以为正或为负 (电感或者电容)



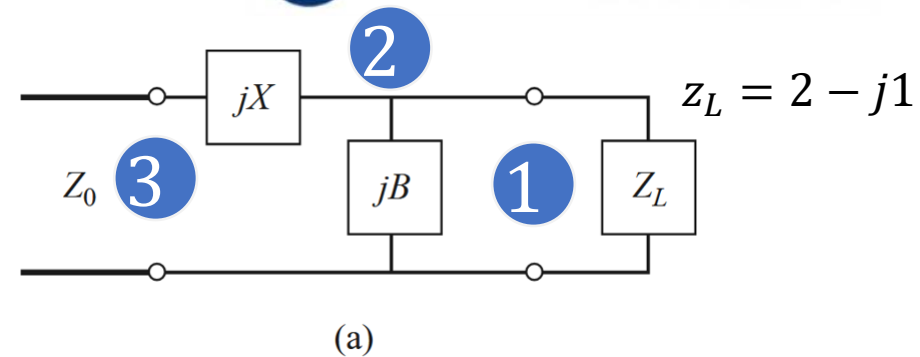
集总元件阻抗匹配-L型匹配



集总元件阻抗匹配

[例] 负载 $Z_L = 200 - j100 \Omega$, 设计一个 L 型匹配网络, 将负载阻抗匹配到 100Ω 的传输线上, 频率为 500 MHz 。

$$z_L = \frac{Z_L}{100} = 2 - j1$$



第一步：1 → 2: 导纳 $+j0.29$

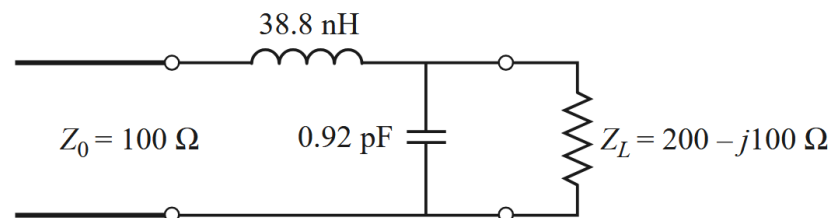
并联电容: $+j\omega CZ_0 = j0.29$

$$C = \frac{0.29}{2\pi \cdot 500e6 \cdot 100} = 0.92 \text{ pF}$$

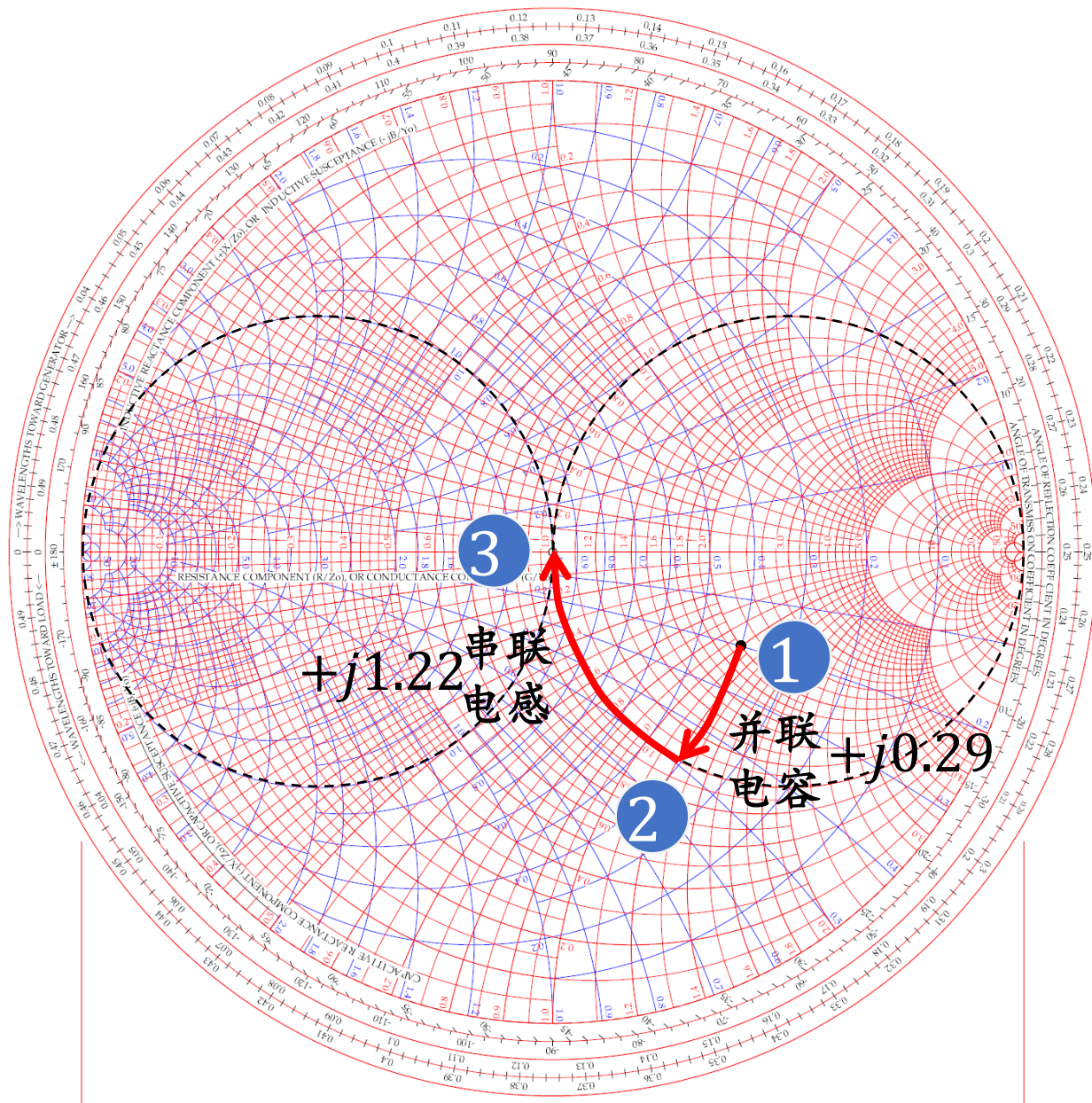
第二步：2 → 3: 阻抗 $+j1.22$

串联电感: $+j\omega L/Z_0 = j1.22$

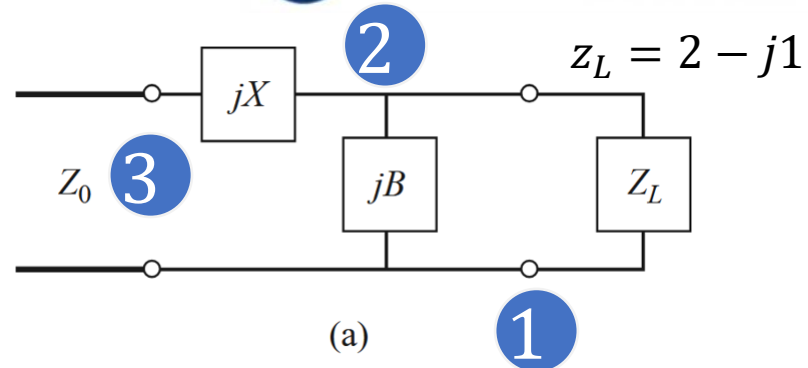
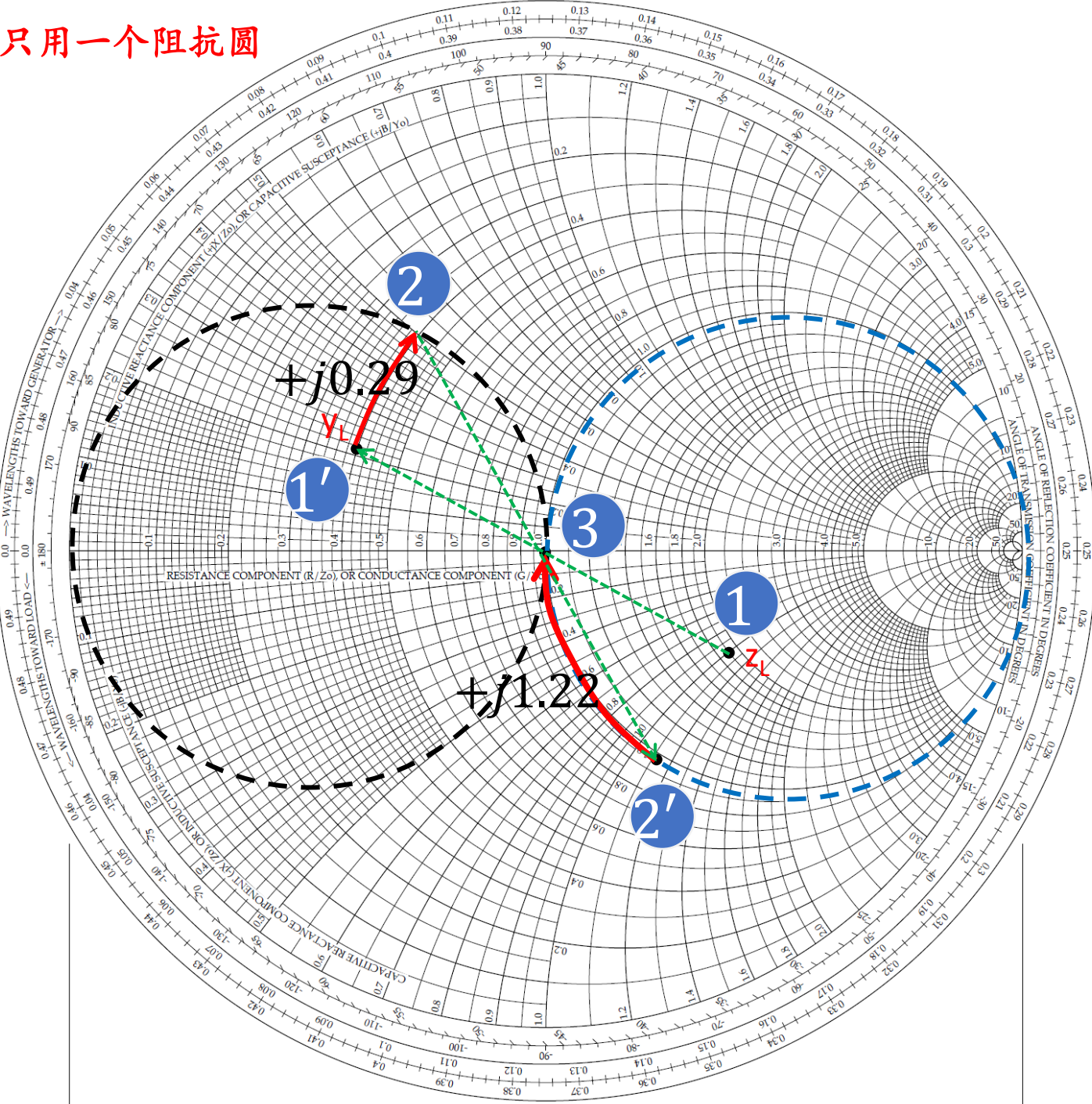
$$L = \frac{1.22 \cdot 100}{2\pi \cdot 500e6} = 38.8 \text{ nH}$$



Solution 1

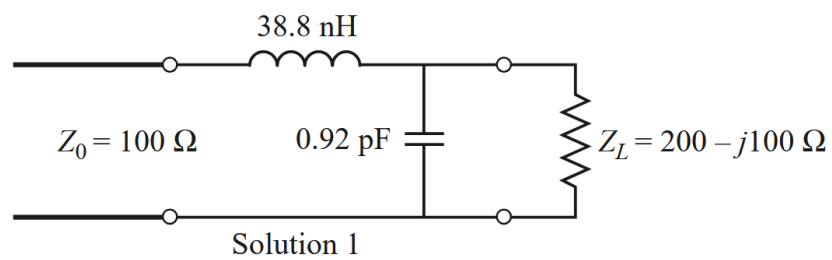


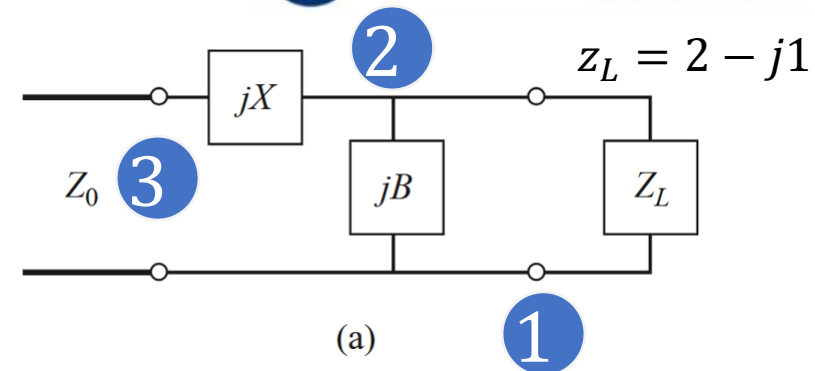
只用一个阻抗圆



第一步：1' → 2: 导纳+j0.29
并联电容： $+j\omega CZ_0 = j0.29$
 $C = \frac{0.29}{2\pi \cdot 500e6 \cdot 100} = 0.92 \text{ pF}$

第二步：2' → 3: 阻抗+j1.22
串联电感： $+j\omega L/Z_0 = j1.22$
 $L = \frac{1.22 \cdot 100}{2\pi \cdot 500e6} = 38.8 \text{ nH}$





第一步：1 → 2: 导纳 $-j0.69$

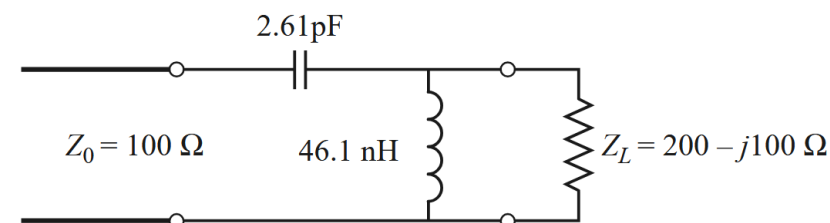
并联电感: $-(jZ_0)/\omega L = -j0.69$

$$L = \frac{100}{0.69 \cdot 2\pi \cdot 500e6} = 46.1 \text{ nH}$$

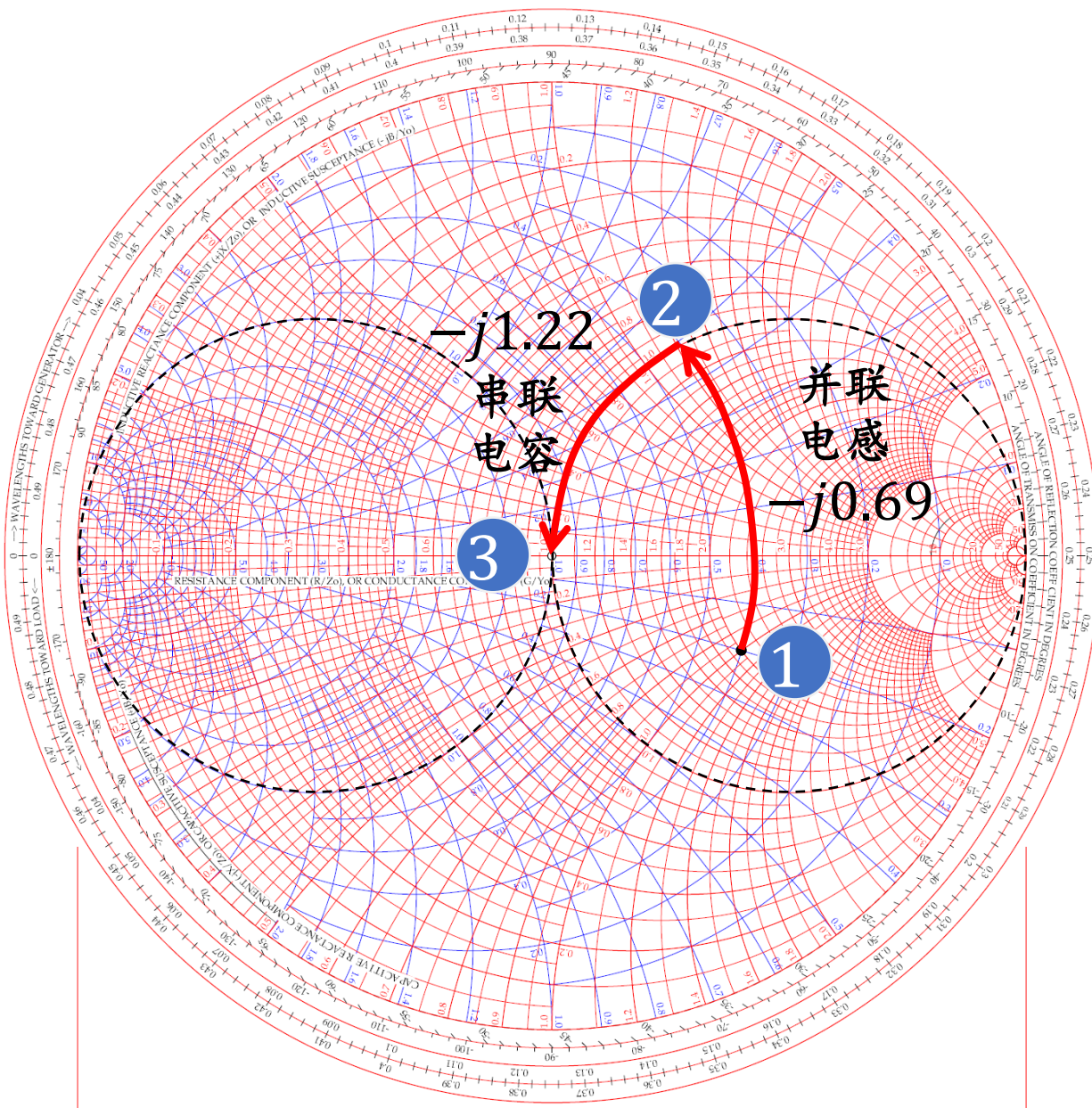
第二步：2 → 3: 阻抗 $-j1.22$

串联电容: $-j1/(\omega CZ_0) = -j1.22$

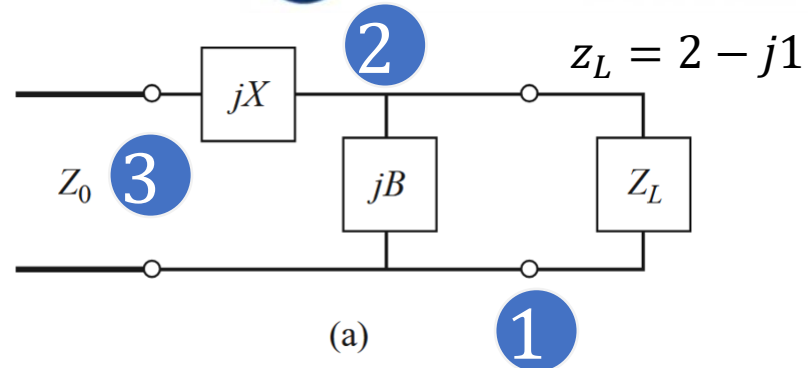
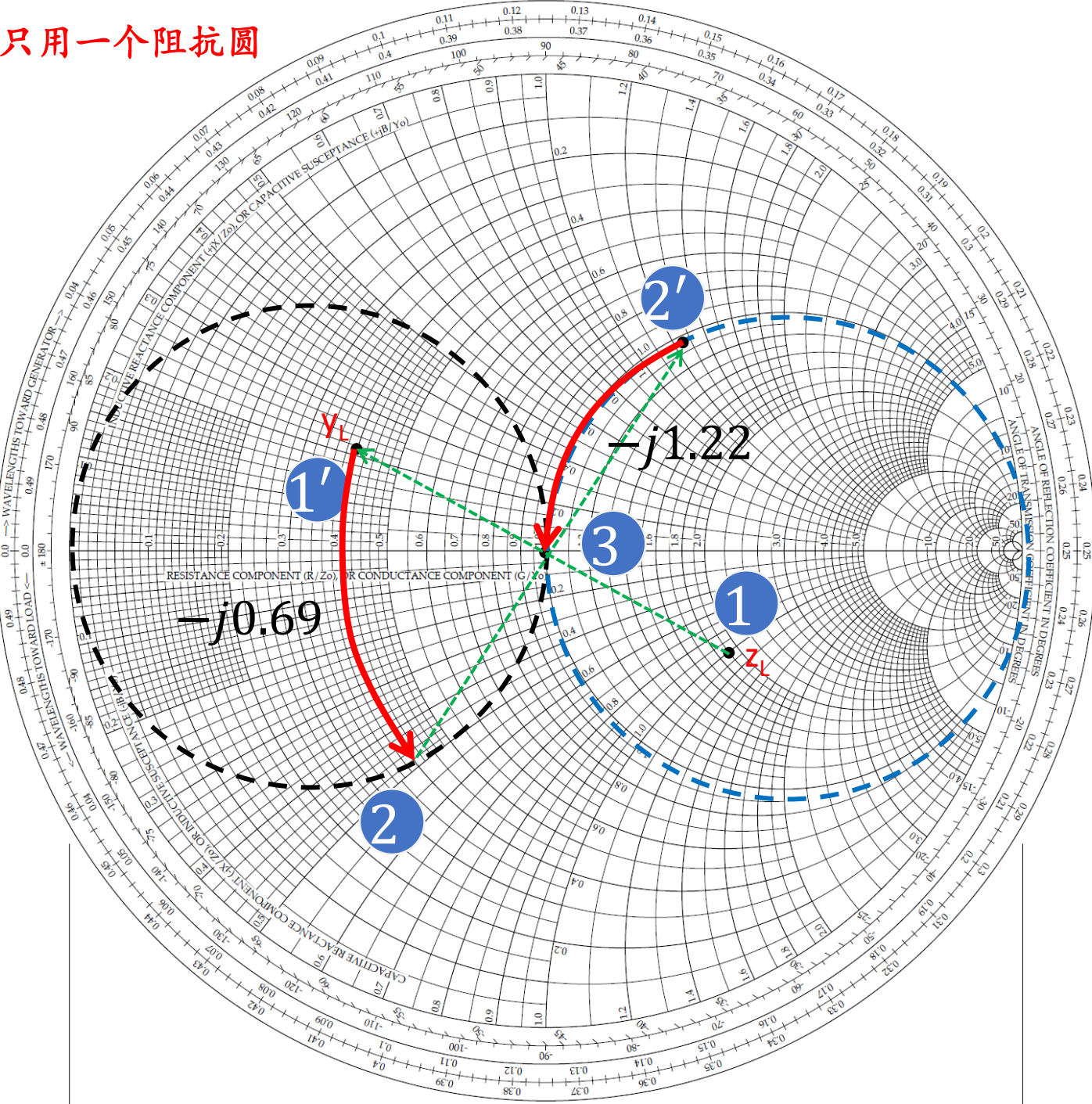
$$C = \frac{1}{1.22 \cdot 2\pi \cdot 500e6 \cdot 100} = 2.61 \text{ pF}$$



Solution 2

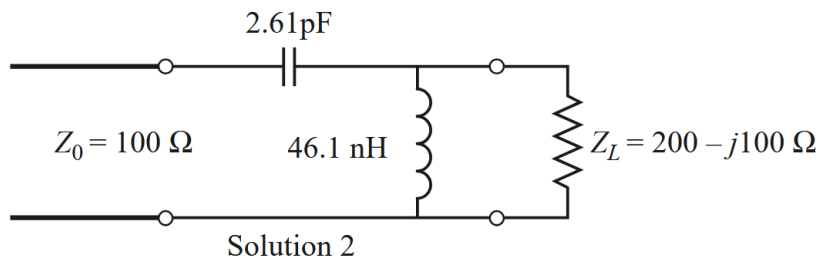


只用一个阻抗圆



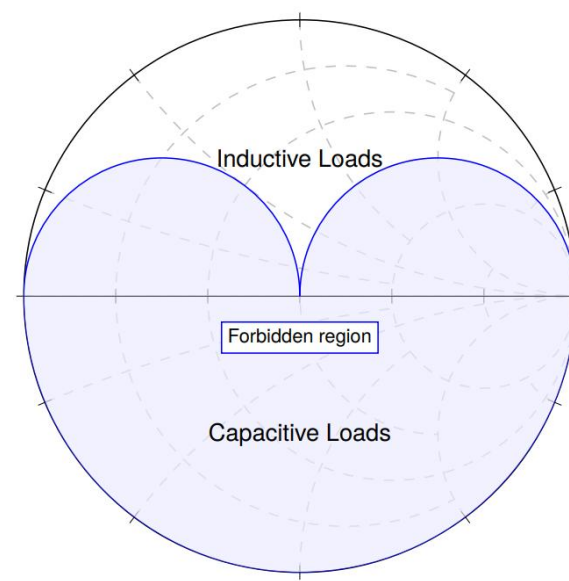
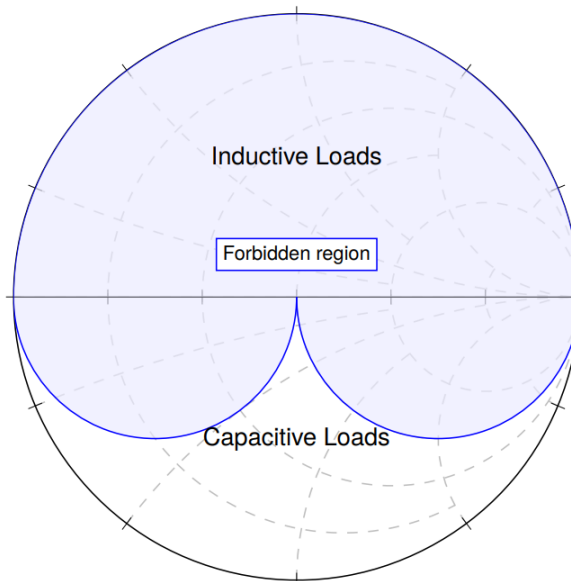
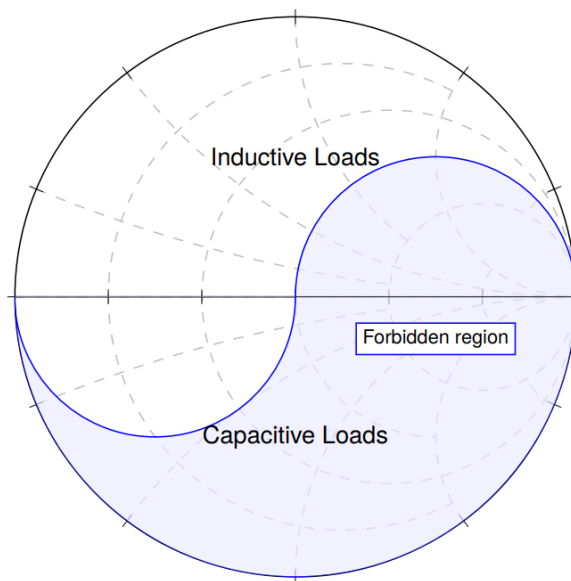
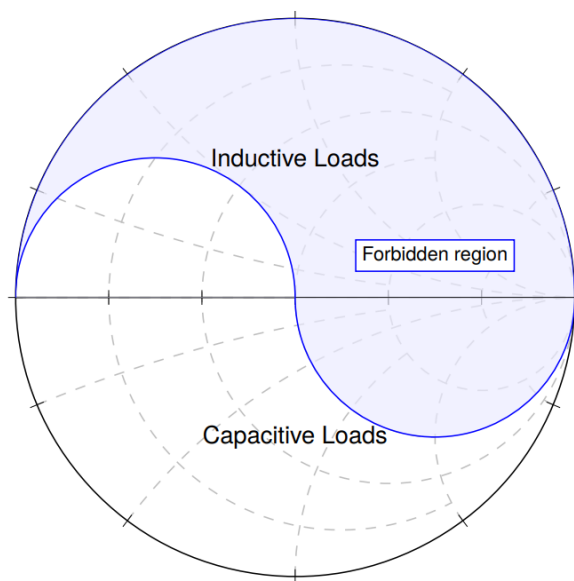
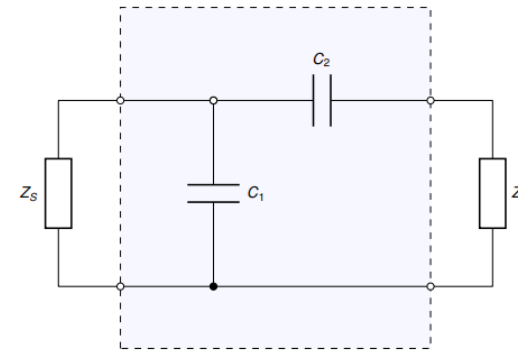
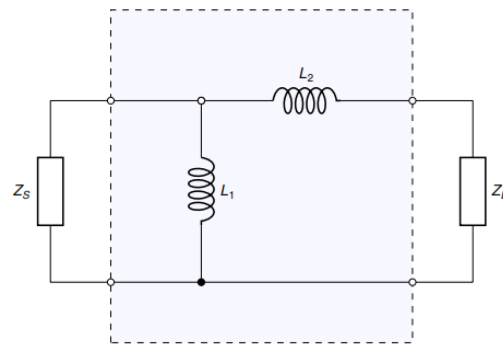
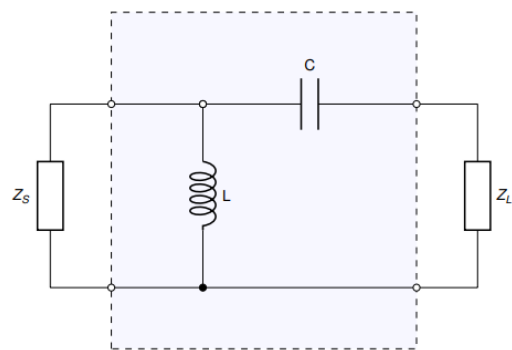
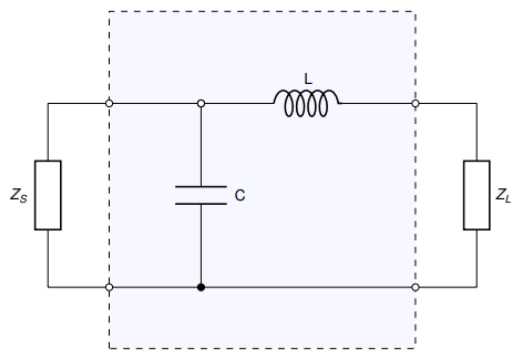
第一步：1' → 2: 导纳-j0.69
并联电感： $-(jZ_0)/\omega L = -j0.69$
 $L = \frac{100}{0.69 \cdot 2\pi \cdot 500e6} = 46.1 \text{ nH}$

第二步：2' → 3: 阻抗-j1.22
串联电容： $-j1/(\omega CZ_0) = -j1.22$
 $C = \frac{1}{1.22 \cdot 2\pi \cdot 500e6 \cdot 100} = 2.61 \text{ pF}$



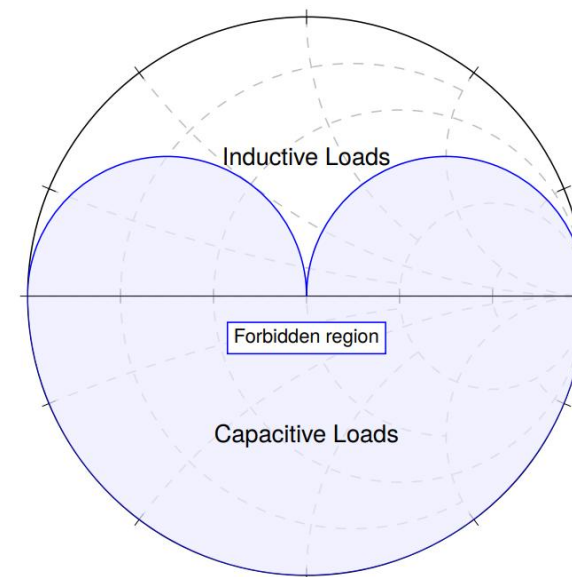
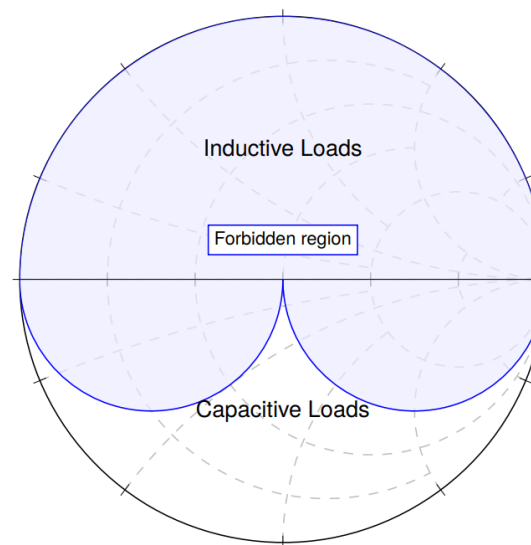
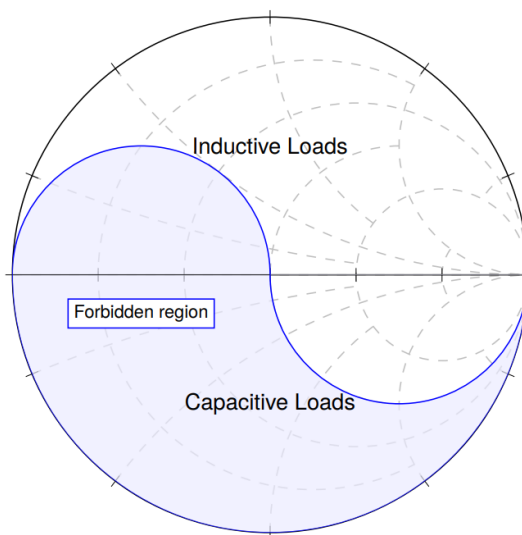
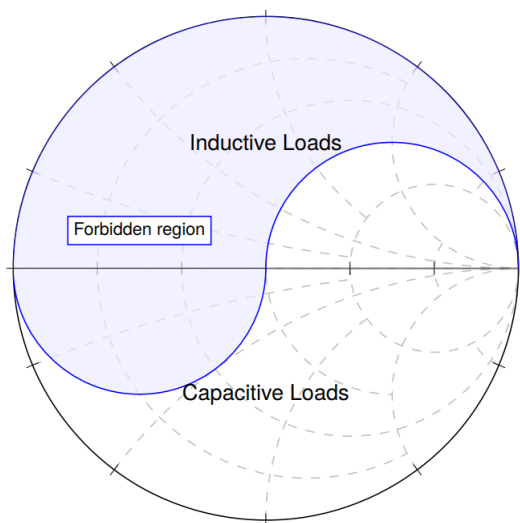
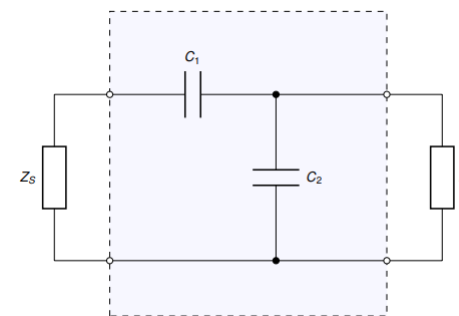
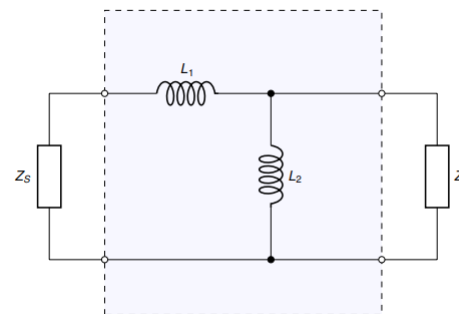
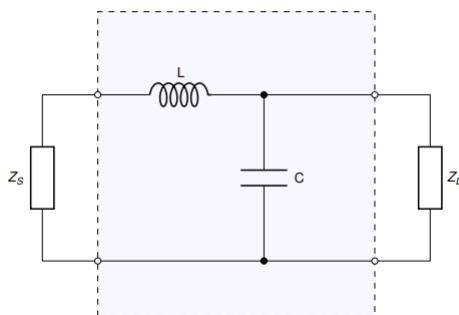
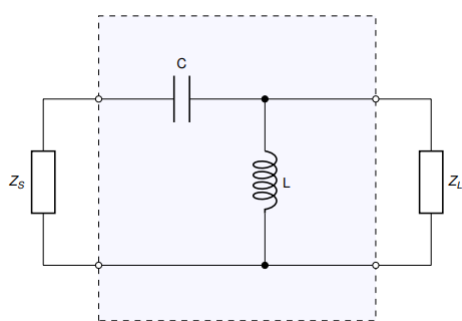
集总元件—L型匹配的匹配禁区

L型匹配电路有8种不同结构，每一种匹配电路都存在难以实现匹配的负载，在史密斯圆图上是一个区域，称为**匹配禁区**。



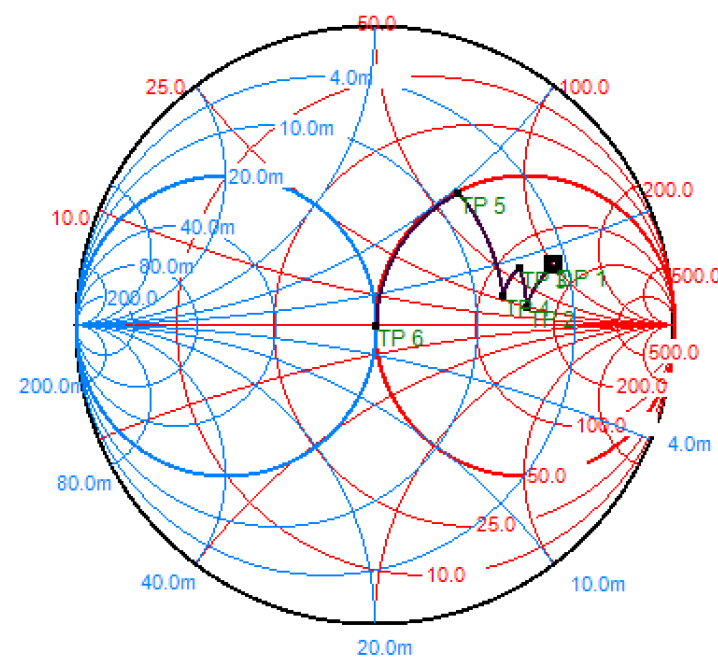
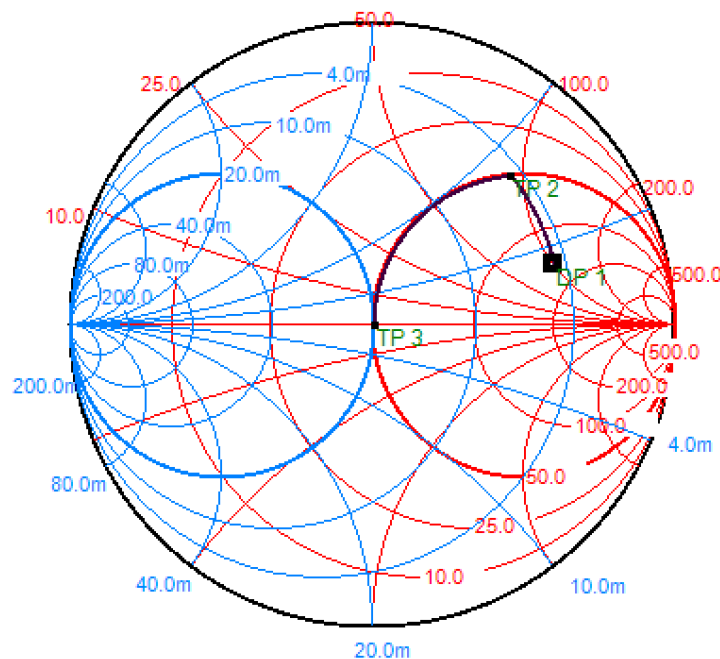
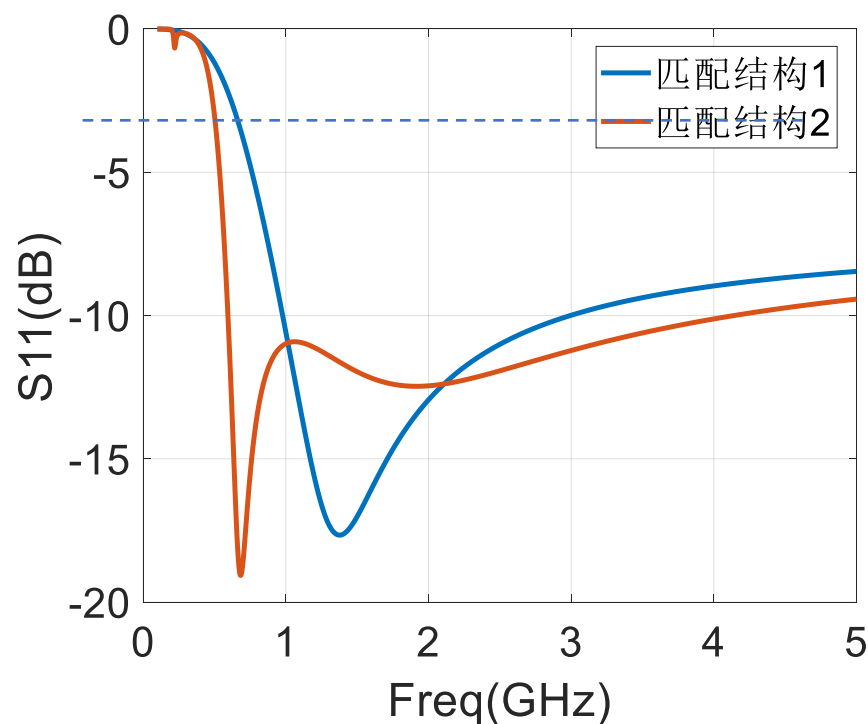
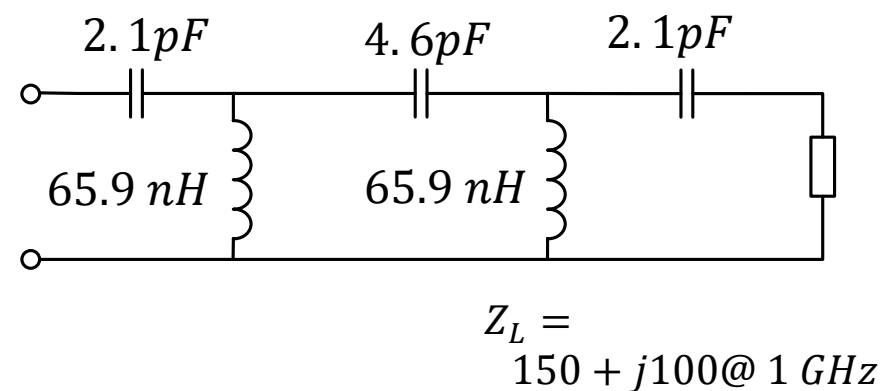
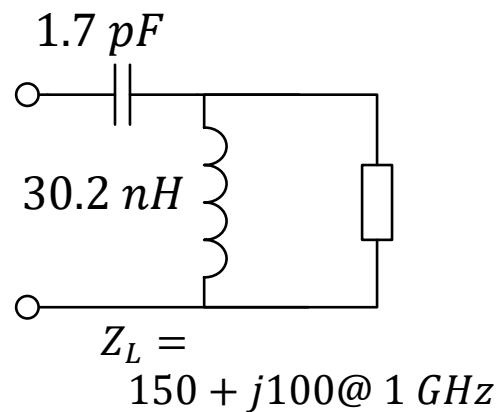
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集总元件阻抗匹配-Q值等值线

- ✓ L型匹配的问题在于带宽窄, 且固定
- ✓ 宽带放大器需要宽带宽
- ✓ 振荡器需要极窄带宽



集总元件阻抗匹配

- 匹配网络可以用有载 Q_L 值描述：

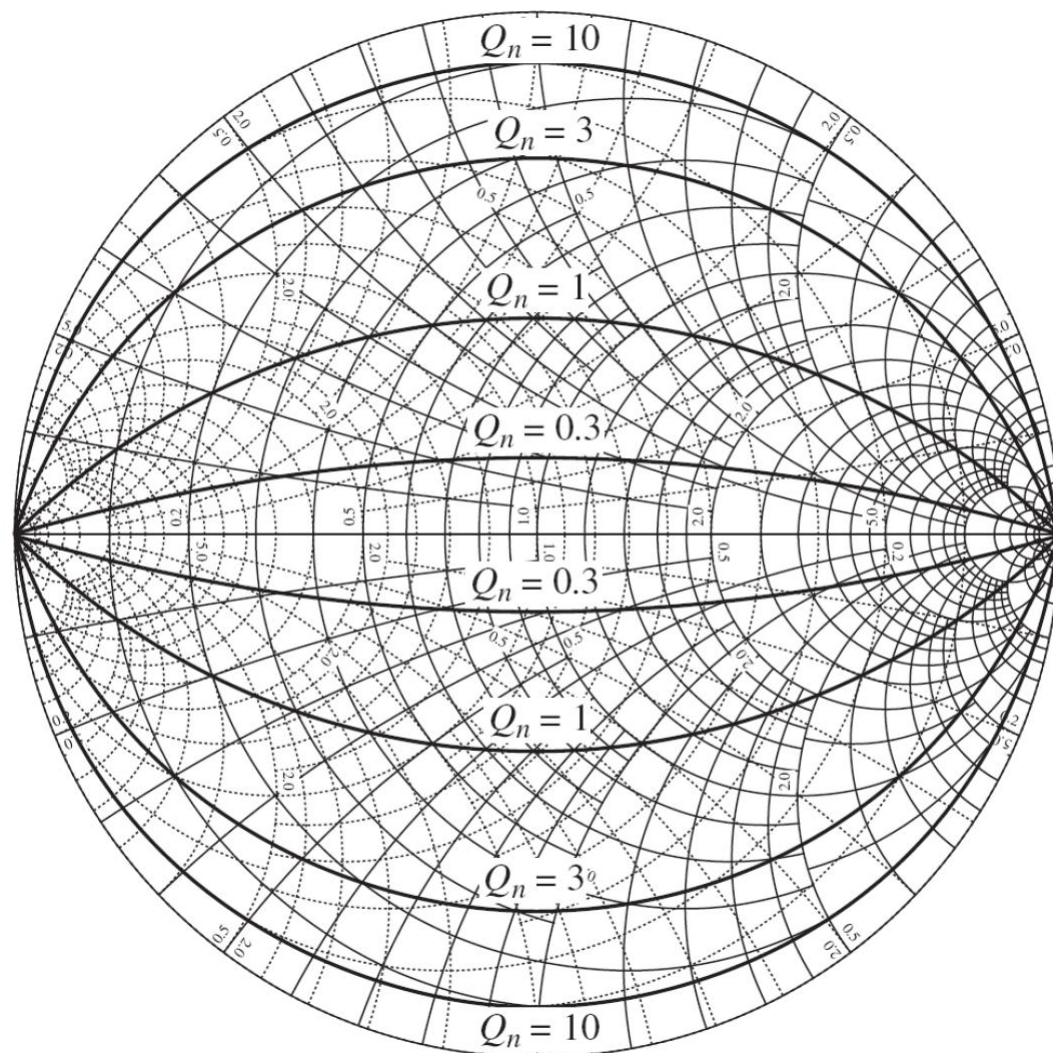
$$Q_L = \frac{f_0}{BW}$$

- Q_L 值描述复杂
- 使用节点 Q 值, Q_n
- 对于任意一个节点, 其值为 $Z_S =$

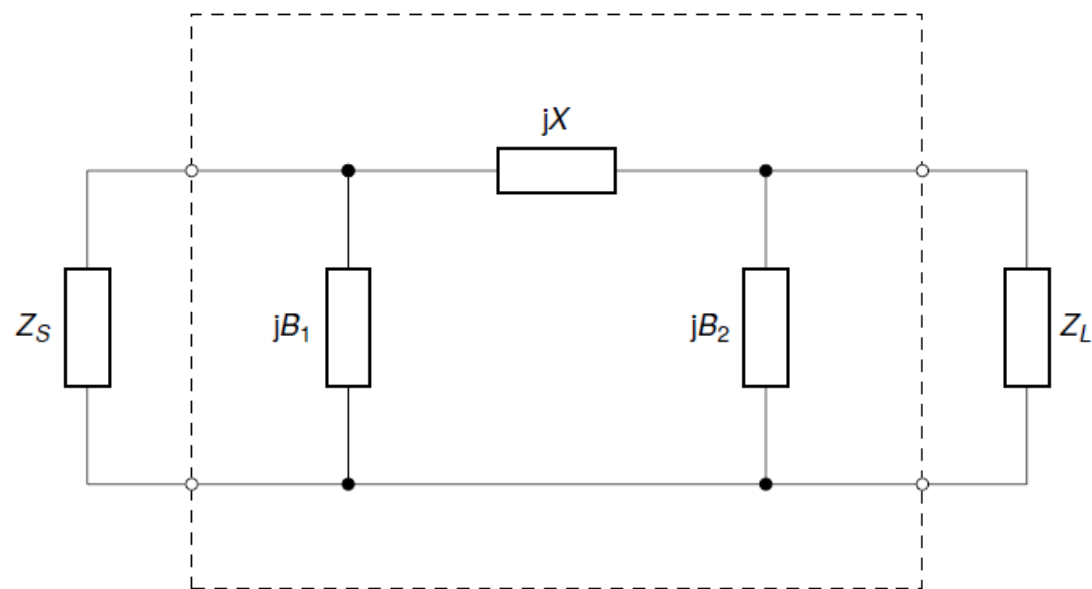
$$R_S + jX_S, \text{ 定义: } Q_n = \frac{|X_S|}{R_S}$$

$$z = r + jx = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} + j \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

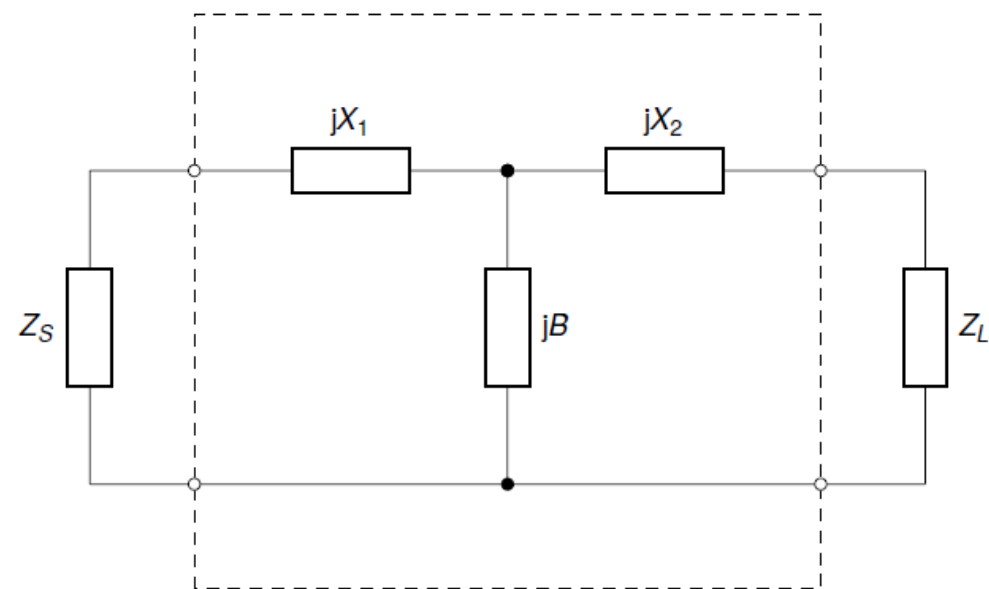
$$\Gamma_i^2 + \left(\Gamma_r \pm \frac{1}{Q_n} \right)^2 = 1 + \frac{1}{Q_n^2}$$



集总元件阻抗匹配-T型/PI型匹配网络

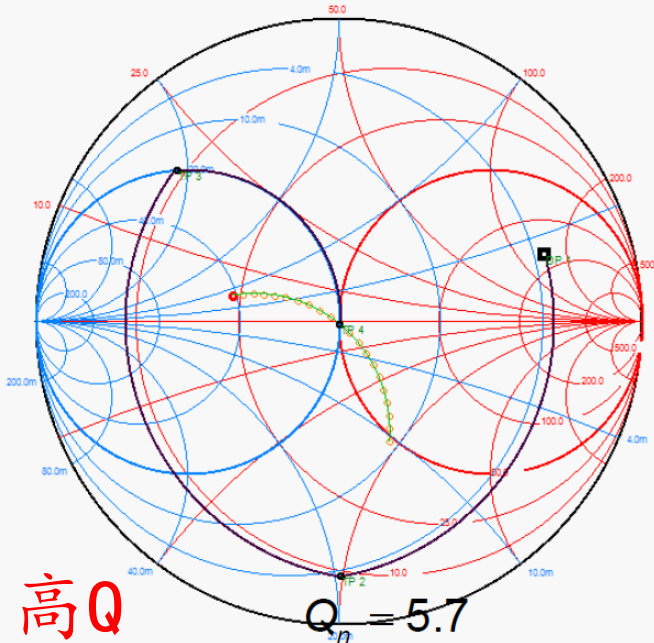
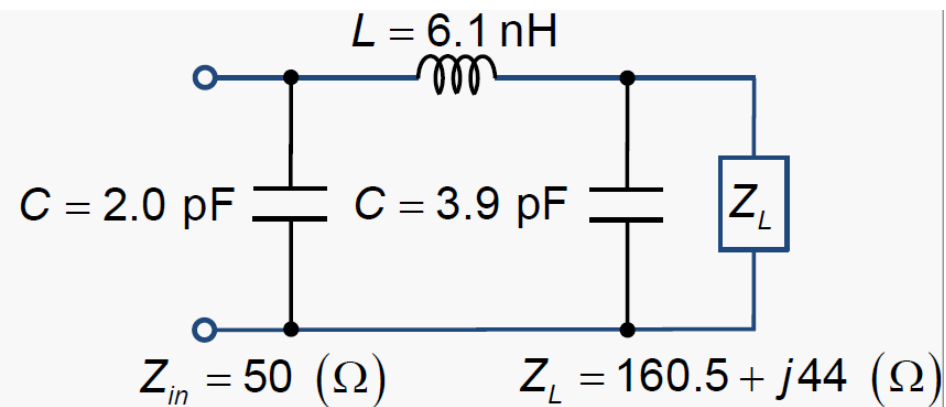
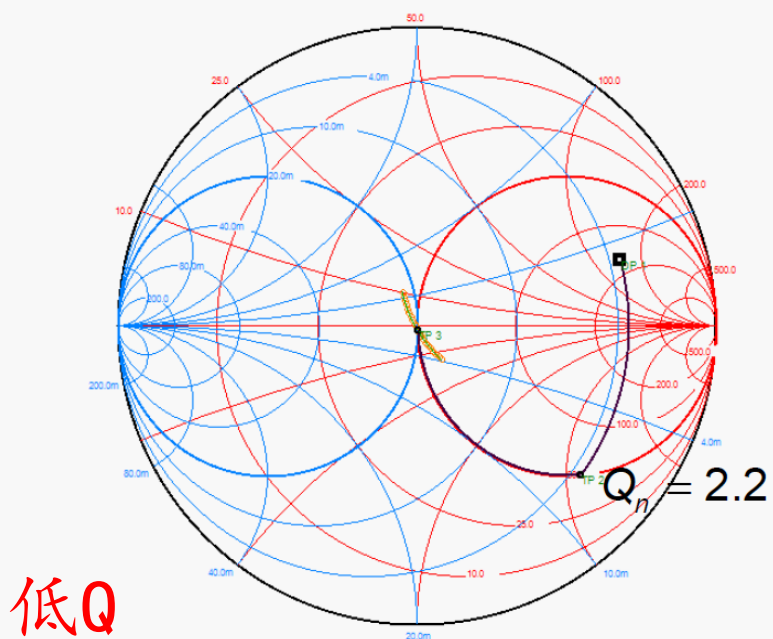
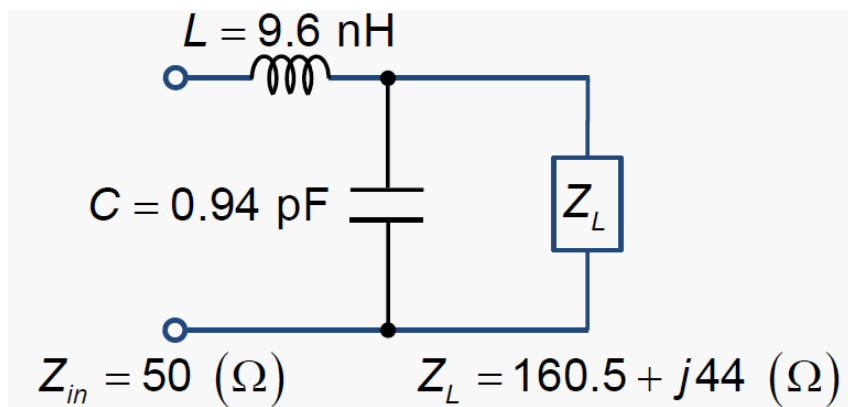


PI-型匹配网络



T-型匹配网络

集总元件阻抗匹配



基于传输线阻抗匹配-单枝节匹配

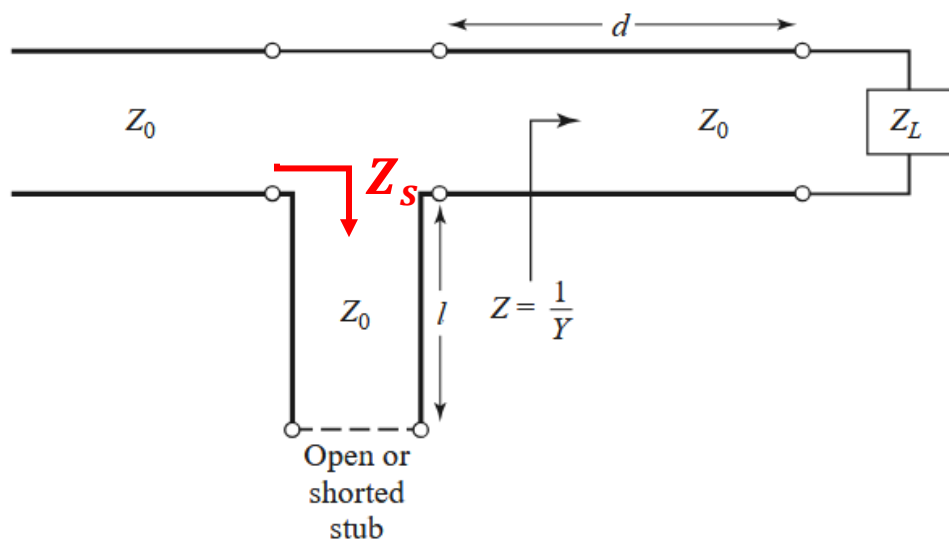
- LC集总元件具有自谐振频率
- 微波LC元件的损耗较大
- LC的元件值固定



无损传输线匹配

$$Z_s = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} = jZ_0 \tan \beta l \quad (\text{Short})$$

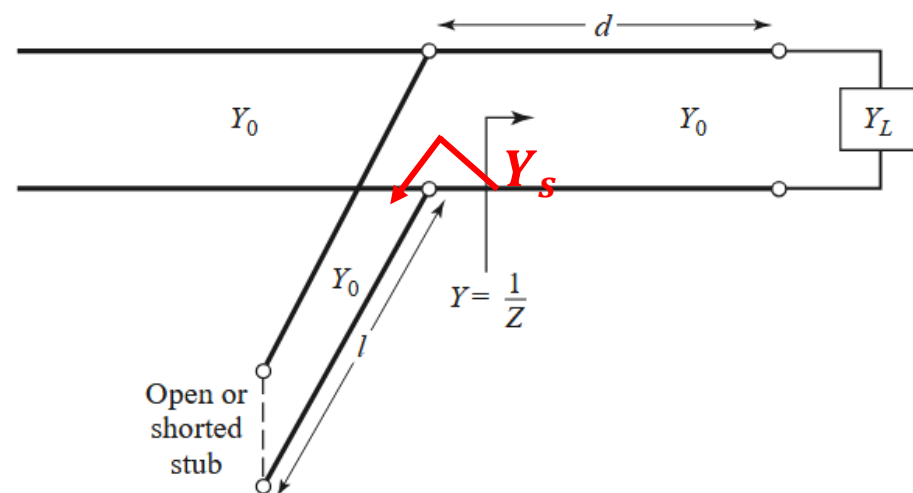
$$= \frac{Z_0}{j \tan \beta l} \quad (\text{Open})$$



串联开/短路枝节匹配

$$Y_s = \frac{1}{jZ_0 \tan \beta l} \quad (\text{Short})$$

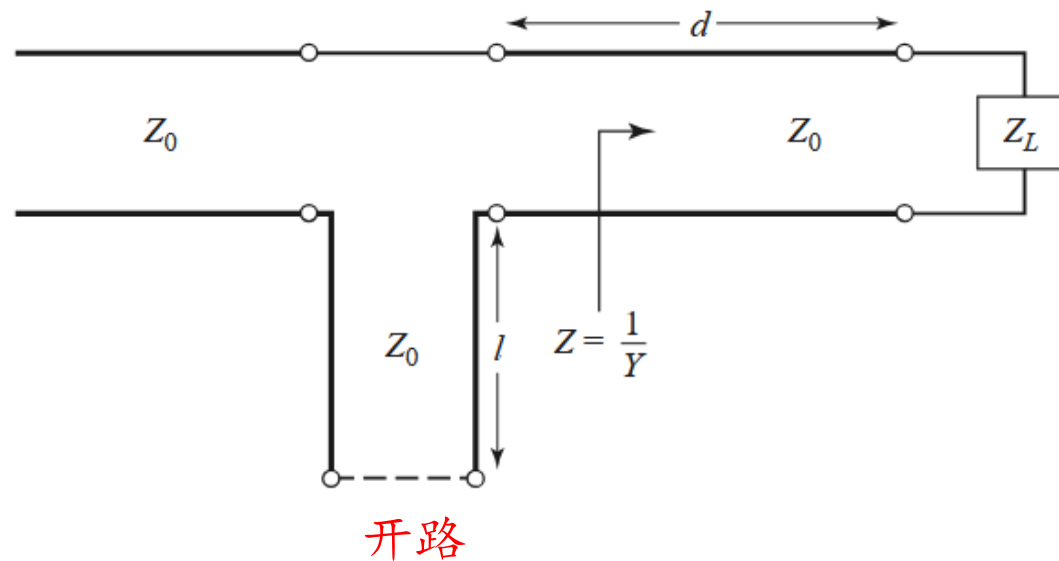
$$= \frac{j \tan \beta l}{Z_0} \quad (\text{Open})$$



并联开/短路枝节匹配

基于传输线阻抗匹配-串联单枝节匹配

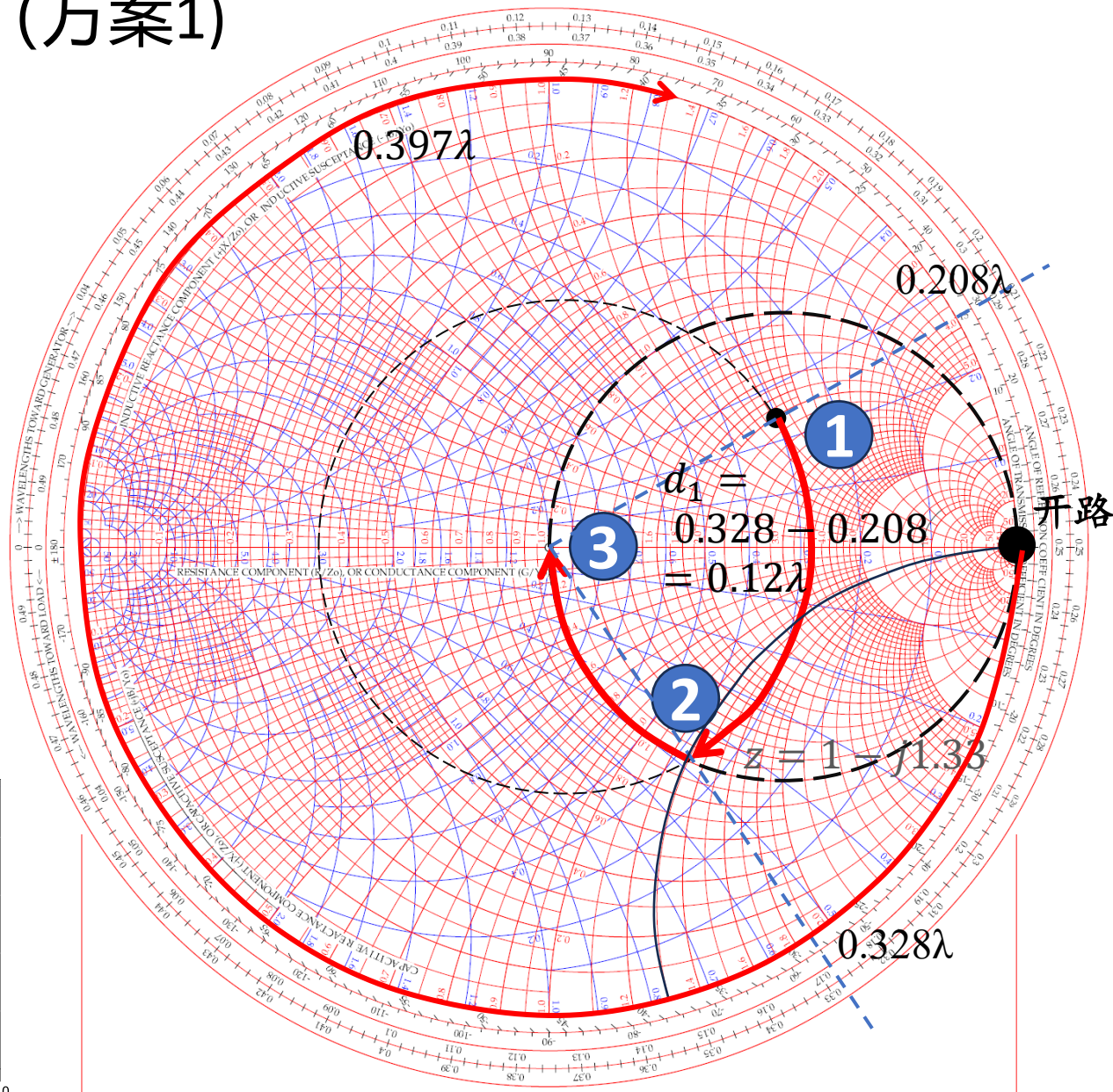
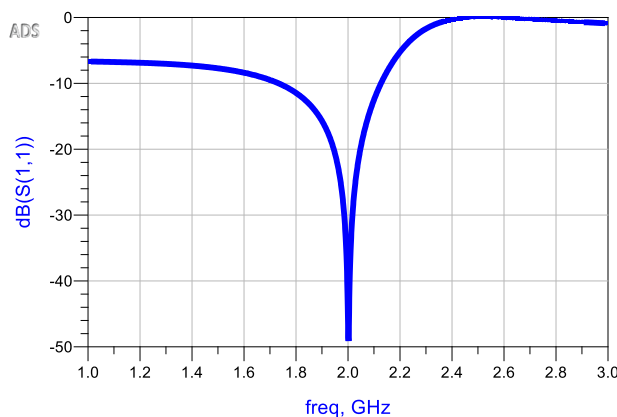
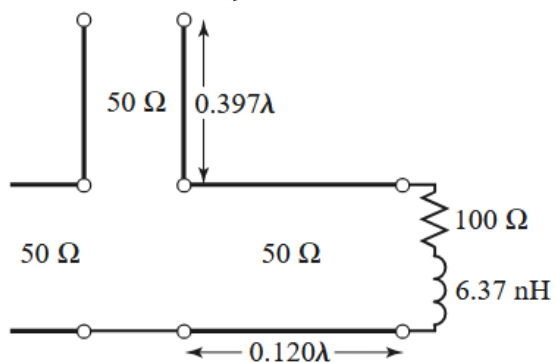
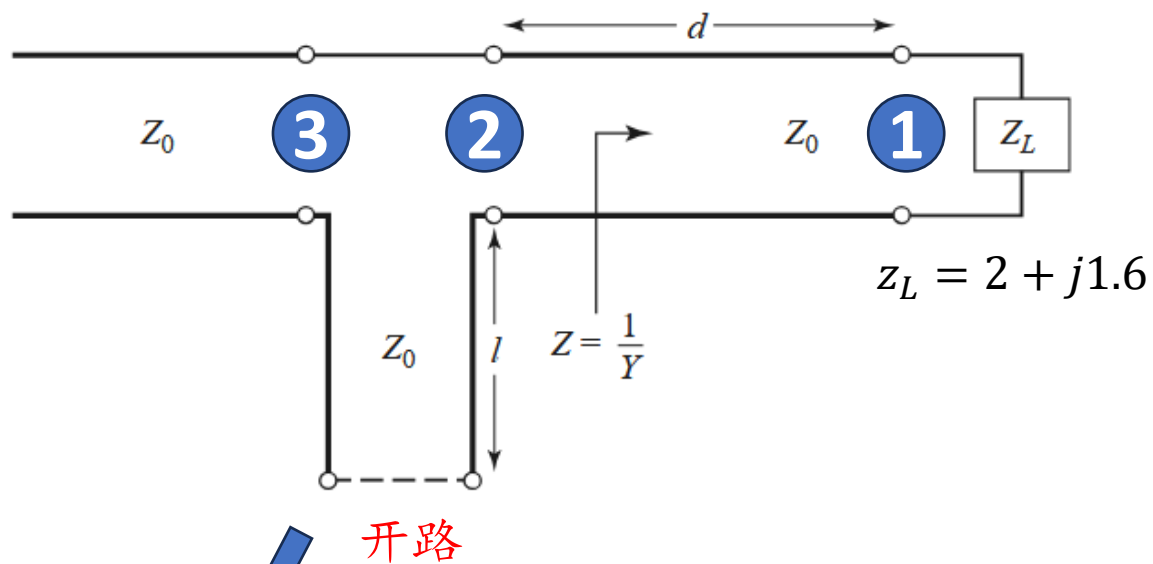
【例题】对于负载阻抗 $Z_L = 100 + j80$ ，设计串联单枝节（开路）网络，将该负载匹配至 $50\ \Omega$ 传输线。假设负载在 $2\ \text{GHz}$ 处匹配，且负载由电阻与电容串联组成，绘制每种解决方案在 $1\ \text{GHz}$ 至 $3\ \text{GHz}$ 频率范围内的反射系数幅度图。



基于传输线阻抗匹配-串联单枝节匹配 (方案1)

$$z = 1 - j1.33$$

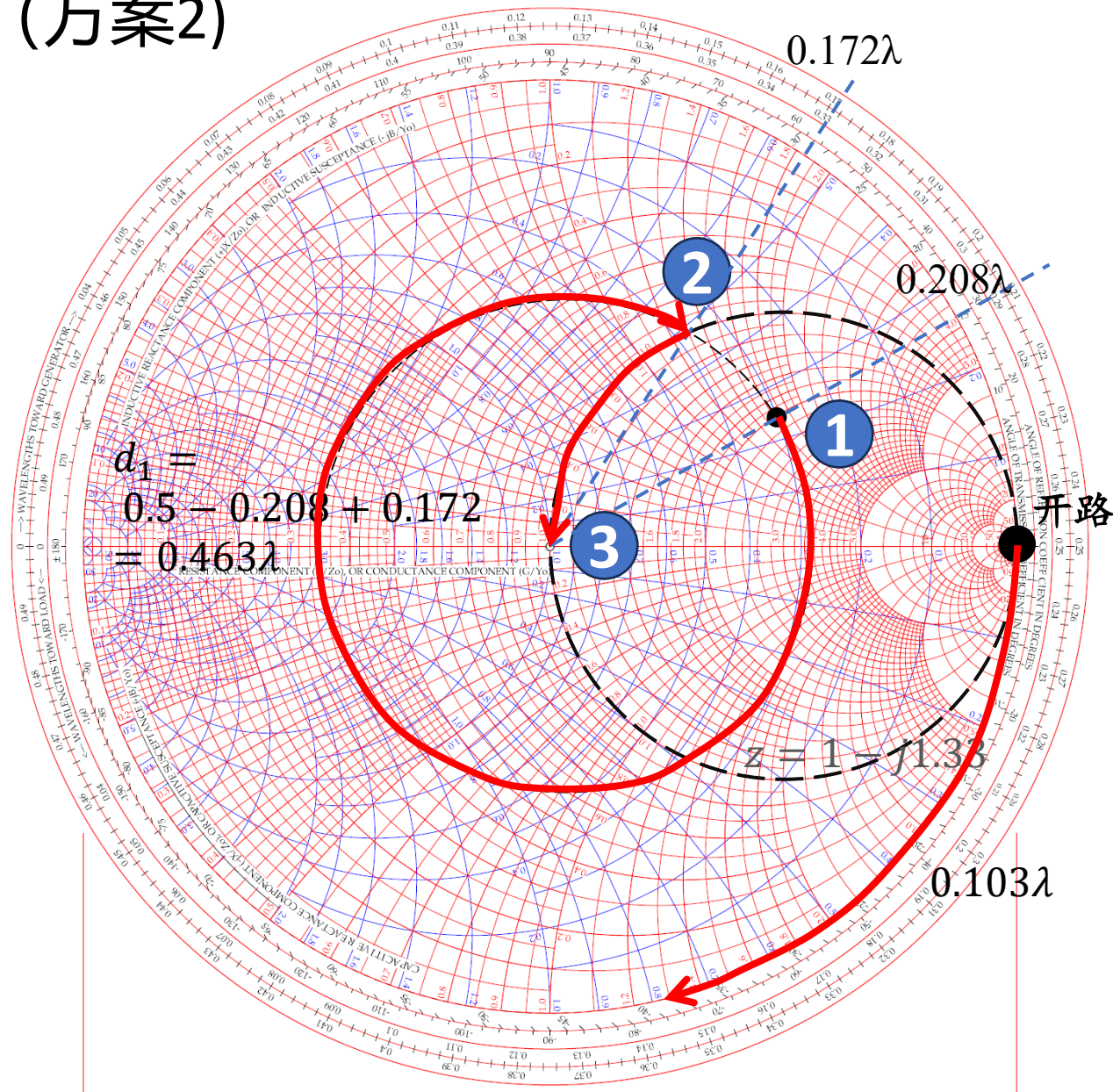
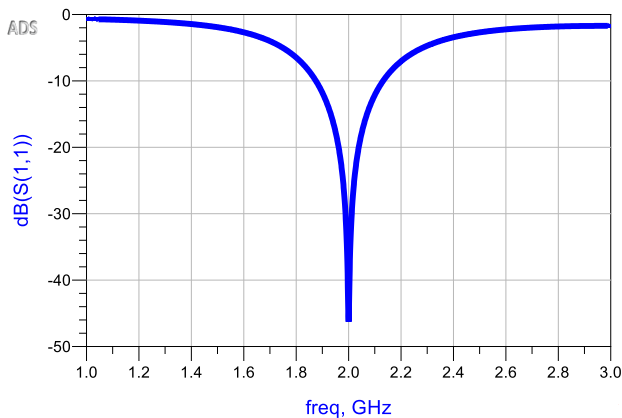
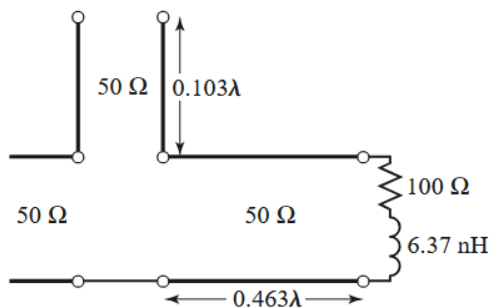
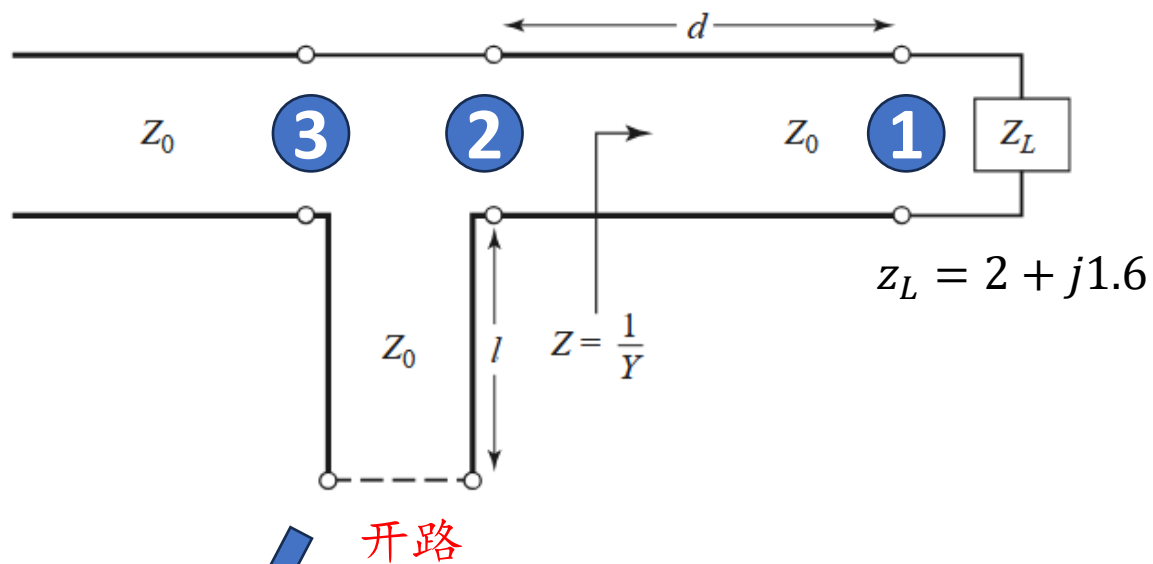
$$\Gamma(l) = \Gamma_L e^{-2j\beta l}$$



基于传输线阻抗匹配-串联单枝节匹配 (方案2)

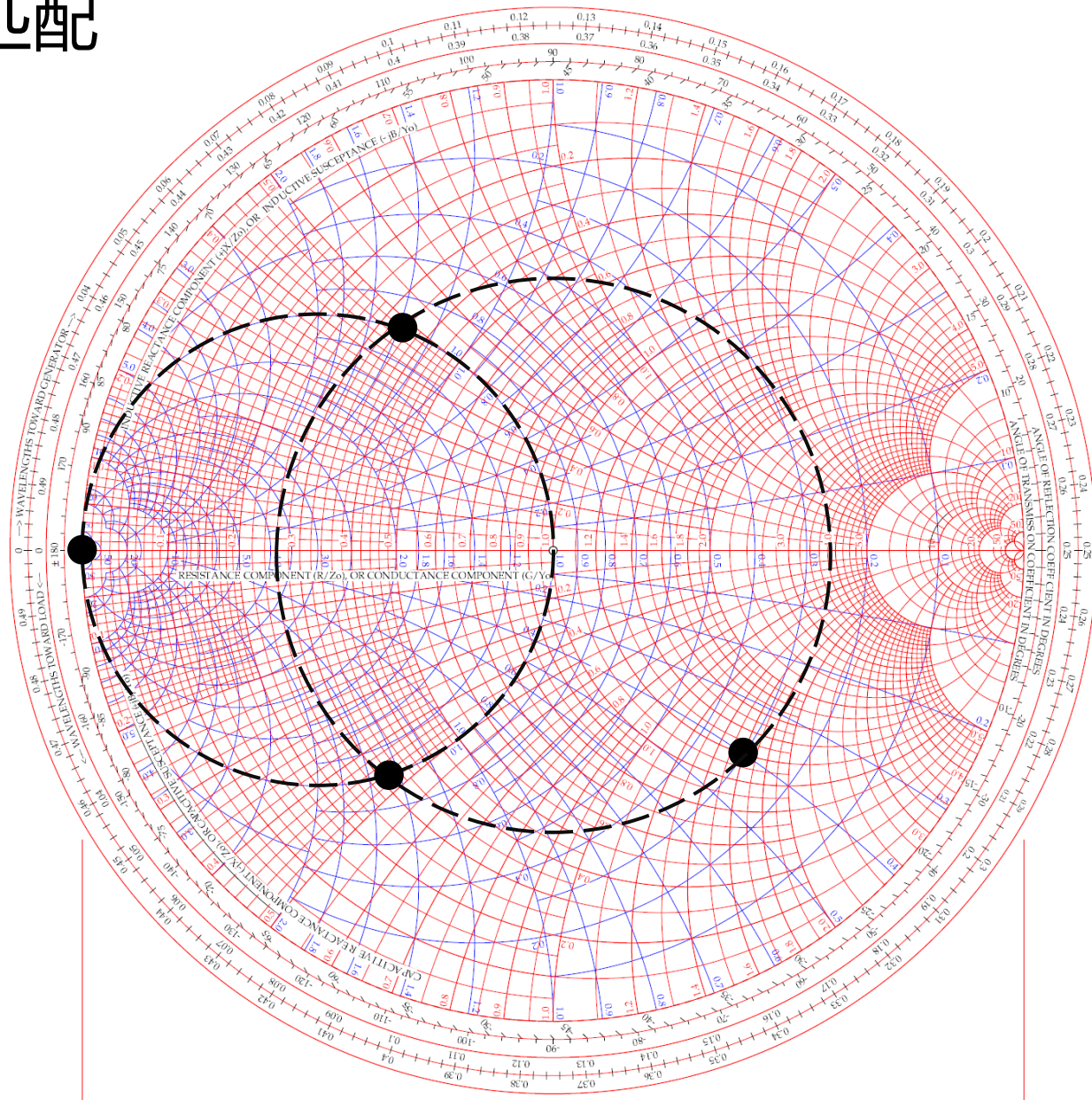
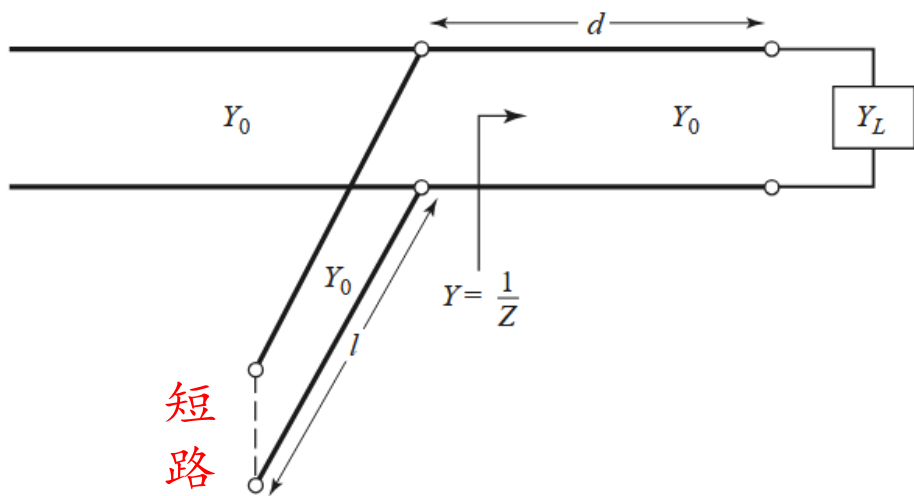
$$z = 1 + j1.33$$

$$\Gamma(l) = \Gamma_L e^{-2j\beta l}$$



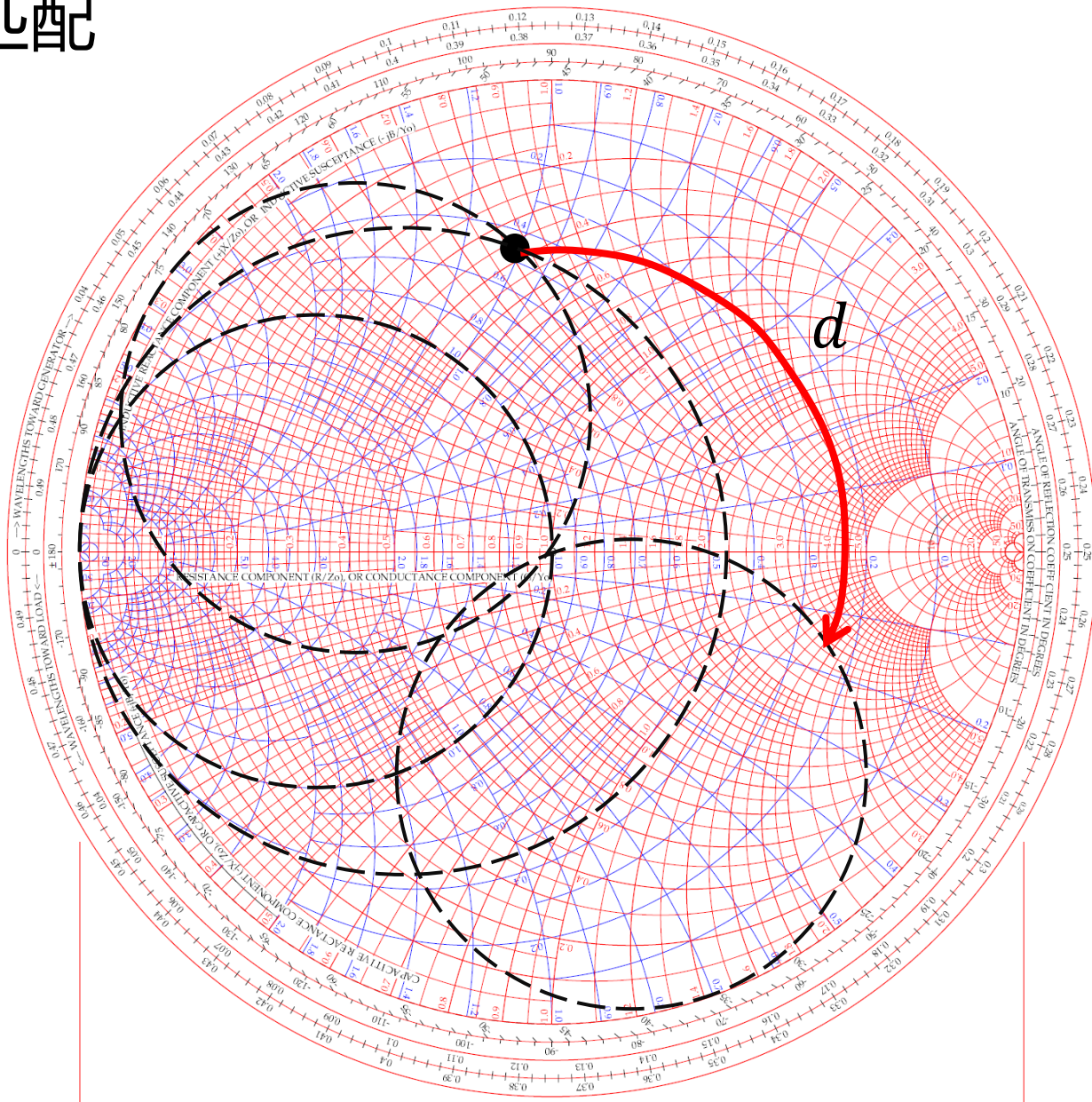
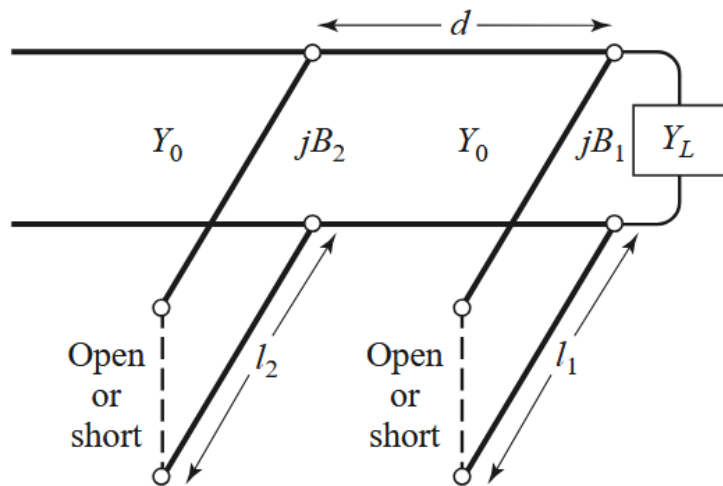
基于传输线阻抗匹配-并联单枝节匹配

【作业】对于负载阻抗 $Z_L = 60 - j80$ ，设计并联单枝节（短路）网络，将该负载匹配至 50Ω 传输线。假设负载在 2 GHz 处匹配，且负载由电阻与电容串联组成，绘制每种解决方案在 1 GHz 至 3 GHz 频率范围内的反射系数幅度图。



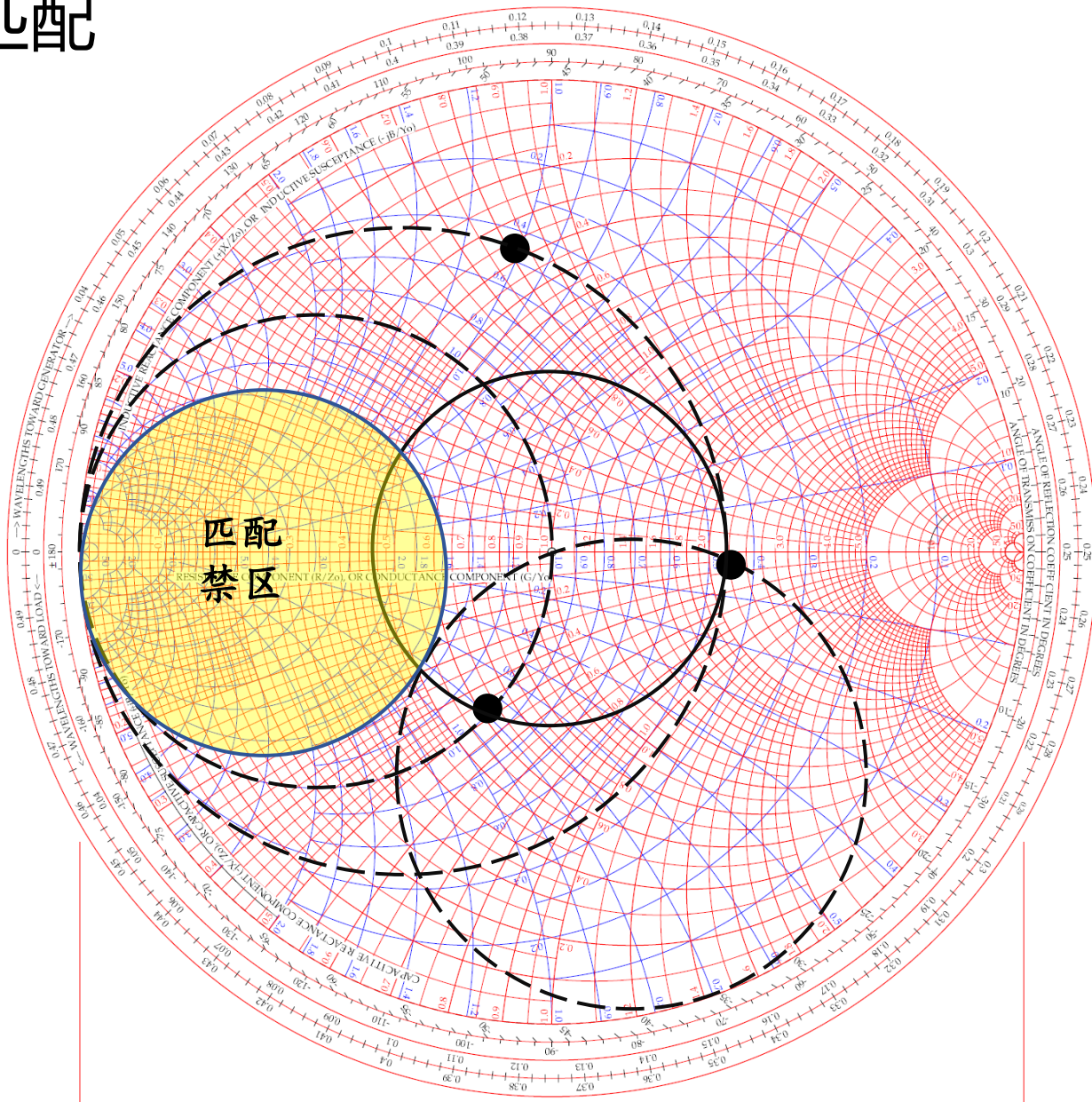
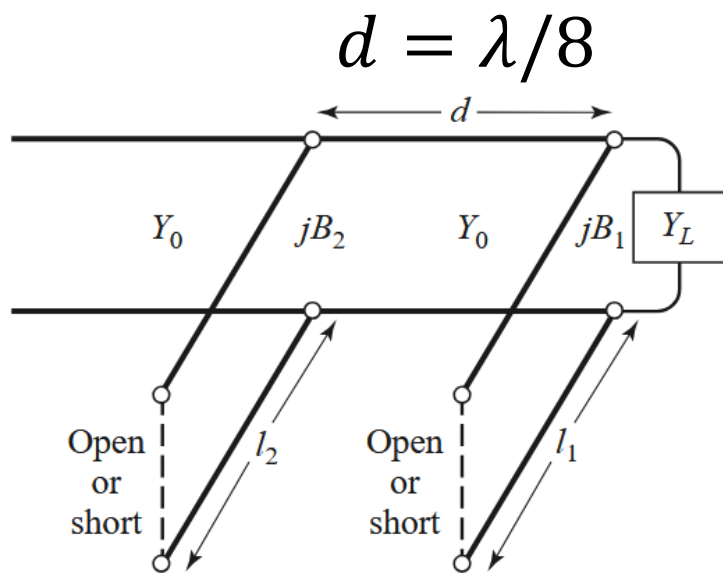
基于传输线阻抗匹配-并联双枝节匹配

- 单枝节匹配的缺点在于长度 d 是随着负载变化的
- 有没有一种方式，传输线的长度不变，而仅需改变枝节的长度即可匹配变化的负载？



基于传输线阻抗匹配-并联双枝节匹配

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- 有没有一种方式，传输线的长度不变，而仅需改变枝节的长度即可匹配变化的负载？



多枝节宽带匹配

